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Programming Exercise

Short Questions

Multiple Choices

Fill in the Blanks

True/False



8.1 Visual Basic Common Controls

Visual Basic provides many controls for creating graphical user interface. Some controls are automatically available in toolbox when Visual Basic starts and some controls need to be added in the toolbox when required. There are two types of controls:

- Intrinsic Controls
- ActiveX Controls

8.1.1 Intrinsic Controls

The controls which are automatically available in toolbox when Visual Basic starts are called **intrinsic controls**. The intrinsic controls are available to every VB program you write.

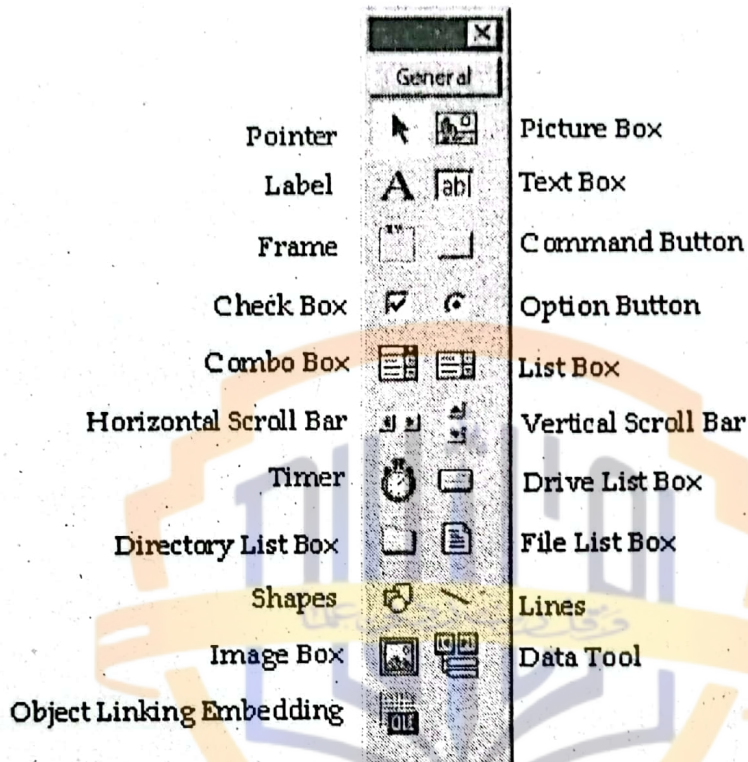


Figure: Intrinsic Controls

8.1.2 ActiveX Controls

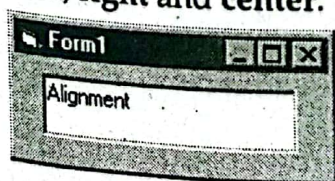
Microsoft uses ActiveX terminology to refer to any control that exists as a separate file with .ocx extension. These controls are available in Visual Basic but not present in the toolbox automatically. These controls need to be included by the user in the toolbox before use.

8.2 Common Properties of Objects

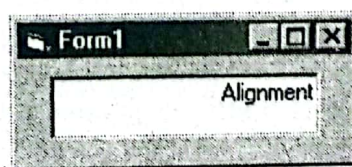
Following properties are found in all objects.

Alignment

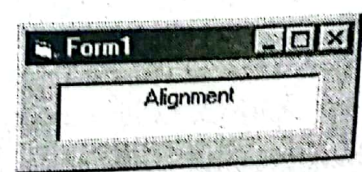
This property is used to align the position of text on the control. Three possible options are left, right and center.



Left alignment



Right alignment



Center alignment

Figure: Alignment property

Appearance

This property is used to set the appearance of the control as follows:

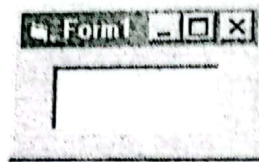


Figure: 1 - 3D

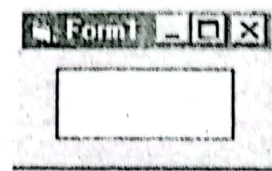


Figure: 0 - Flat

BackColor

This property is used to set the background color of the control. You can select a color from color palette or give a hexadecimal value of the color.

BorderStyle

This property is used to set the style of border of the control. You can select a style from a predefined list of styles.

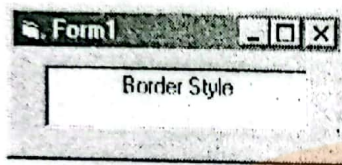


Figure: 3-D border style

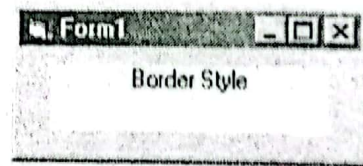


Figure: Flat border style

Caption

This property is used to assign the text displayed on the buttons and labels etc.

Enabled

This property is used to enable or disable a control. It is either **True** or **False**. If false, the text on the control is dimmed.

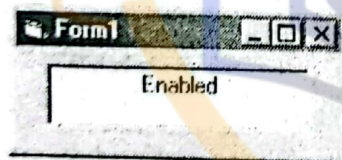


Figure: Enabled property True

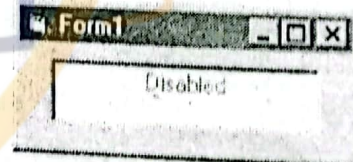


Figure: Enabled property False

Font

This property is used to set font, font style and size of the text displayed on the control.

ForeColor

This property is used to set the foreground color of the control. It is set either by selecting a color from color palette or by giving hexadecimal value of the color.

Height

This property is used to set the height of a control in twips.

Left

This property is used to set the distance of the control from left corner of the form. In case of form, Left is used to set the left distance from screen.

MousePointer

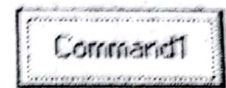
This property is used to set the shape of mouse when mouse is moved over the control.

TabIndex

This property is used to set the tab index of controls. It is used when TAB key is pressed at runtime. For example, if the control is on a control with TabIndex 0, the control will move to the control with TabIndex 1 when TAB is pressed.

ToolTipText

This property is used to set the text that will appear when the user stops the mouse over a control.



Hello

Top

This property is used to set the distance of the control from top corner of the form. In case of form, Top is used to set the left distance from screen.

Visible

This property is used to hide or unhide a control. It is either True or False. If False, the control is not visible during execution.

Width

This property is used to set the width of a control in twips.

8.3 Command Button Control

Command button is very important and commonly used control in Visual Basic. The user mostly interacts with the program by clicking on different buttons.

8.3.1 Important Properties of Command Button

Some important properties of command button are as follows:

Picture

This property is used to display picture on the button. The picture will appear only if the Style property is set to Graphical.

Style

This property is used to set the style of the button.

Option	Description
0 – Standard	The default style.
1 – Graphical	Used to display color or picture on the button.

8.3.2 Important Events of Command Button

Some important events of command button are as follows:

Click

The Click event occurs when the user clicks on the button.

DbClick

The DbClick event occurs when the user double clicks on the button.

MouseMove

The MouseMove event occurs when the user move mouse over the button.

GotFocus

The GotFocus event occurs when the button gets focus.

LostFocus

The GotFocus event occurs when the button loses focus.

8.3.3 Accessing Button from Keyboard

A button can be accessed from keyboard in two ways:

- Press TAB key to move the focus on the button and then press SPACE key.
- Assign a key that will be used along with of ALT key to access that button.

The ampersand & is used to assign a key. If ampersand is used before any letter in Caption property, that letter will be underlined. At runtime, the user can press that letter from keyboard while holding ALT key to access the button.

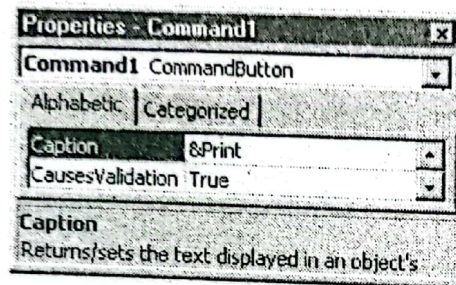


Figure: Caption property



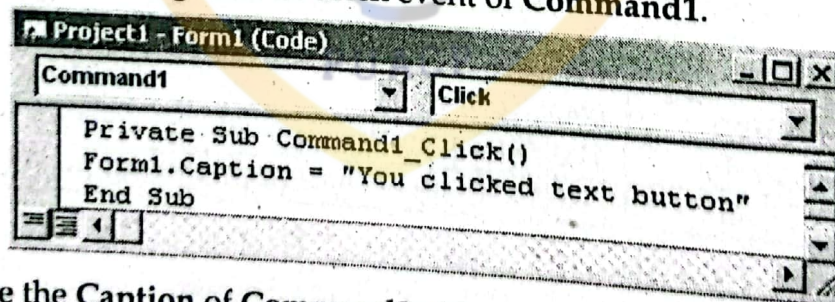
Figure: Effect on the button

The user can press ALT+P to activate the above command button.

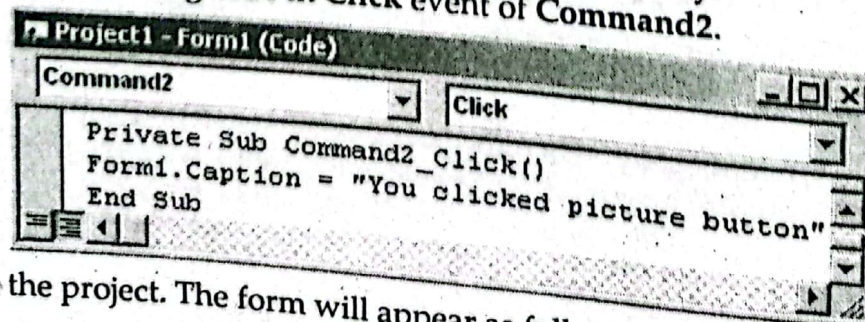
Program 8.1

Write a program that displays two command buttons on a form. The first button has caption "Click Me" and the other has a picture on it. The caption of form should change if the user clicks a button.

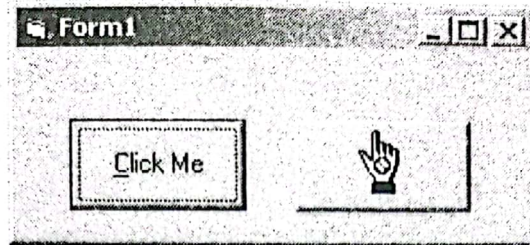
1. Start a new Standard EXE project.
2. Place two buttons on the form.
3. Set Caption of Command1 to "&Click Me".
4. Add the following code in Click event of Command1.



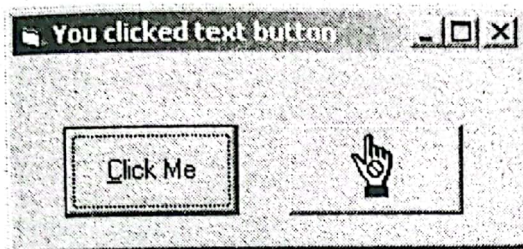
5. Delete the Caption of Command2 and set its Style property to Graphical.
6. Select any icon for Command2 from Picture property.
7. Add the following code in Click event of Command2.



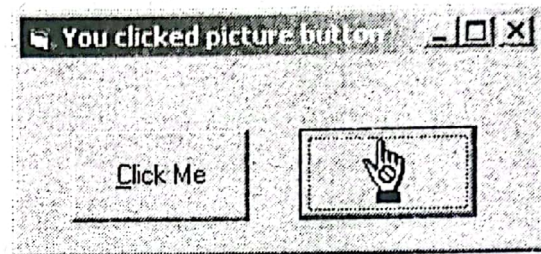
8. Run the project. The form will appear as follows:



9. Click the buttons. The caption of the form will change as follows:



Command1 clicked



Command2 clicked

8.4 Textbox Control

Textbox control is one of the most important controls in Visual Basic. It is mostly used to get the input of the user.

8.4.1 Important Properties of Textbox Control

Some important properties of Textbox control are as follows:

Locked

This property is a **Boolean property**. If it is **True**, the user cannot change the contents of the textbox during execution. The default setting is **False** and the user can enter and change the contents of a textbox.

MaxLength

This property is used to set the **maximum number of character** that can be entered by the user in textbox. The value 0 indicates **unlimited number of characters**.

MultiLine

This property is a Boolean property. If it is **True**, the user can enter text in multiple lines. The default setting is **False** and the text is entered in single line.

PasswordChar

This property is used to set the password character for the textbox. When the user enters any text, the given character will appear in the textbox instead of the text entered by the user.

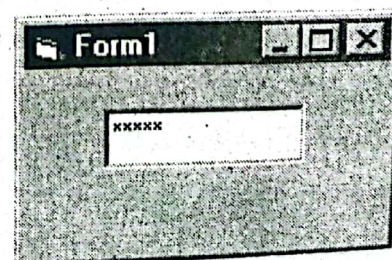
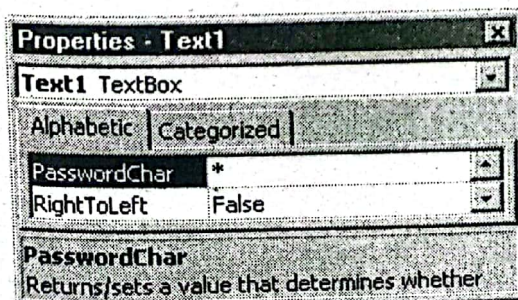


Figure 8.4.1 PasswordChar property

RightToLeft

This property is a Boolean property. If it is True, the text in the textbox will appear from right to left. The default setting is False and the text appears from left to right side.

ScrollBar

This property is used to show scrollbars with the textbox. The scrollbars will appear only if the **MultiLine** property of the textbox is True.

Option	Description
0 -- None	No scrollbar appears
1 -- Horizontal	Only horizontal scrollbar appears
2 -- Vertical	Only vertical scrollbar appears
3 -- Both	Both horizontal and vertical scrollbars appear

Table: Scrollbar options

SelLength

This property is used to get the number of characters selected by the user in the textbox. This property can be used only at runtime.

SelStart

This property is used to get the position of the character from where the selection starts. This property can be used only at runtime.

SelText

This property is used to get the selected text. It can be used only at runtime.

Tag

This property is used to store text related to textbox. This text can be stored for different purposes during execution. You can store the password in this property etc.

Text

This property represents the contents of the textbox.

8.4.2 Important Methods of Textbox Control

Some important methods of Textbox control are as follows:

SetFocus

It is used to set the focus to the textbox. The syntax of using this method is as follows:

Text1.SetFocus

8.4.3 Important Events of Textbox

Some important events of textbox are as follows:

Change

This event occurs when user changes the contents of textbox during execution.

GotFocus

This event occurs when the control move to the textbox.

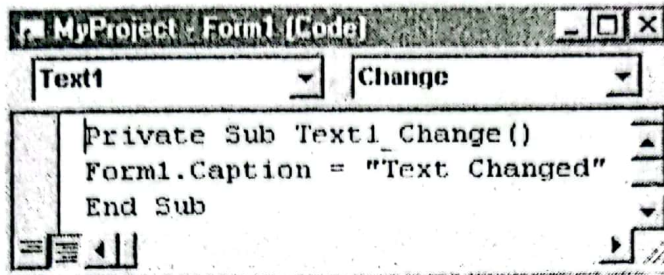
LostFocus

This event occurs when the controls leaves the textbox.

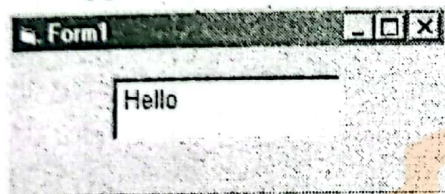
Program 8.2

Write a program that changes the caption of the form to "Text Changed" when the user makes any change to the contents of a textbox.

1. Start a new Standard EXE project.
2. Place a textbox on the form and set its Text property to "Hello".
3. Add the following code in the Change event of the textbox.



4. Run the project.
5. Type any text in the textbox. The caption of the form will change.



Caption before change

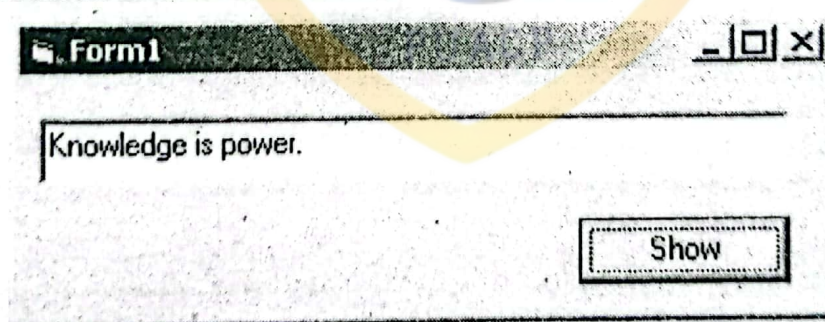


Caption after change

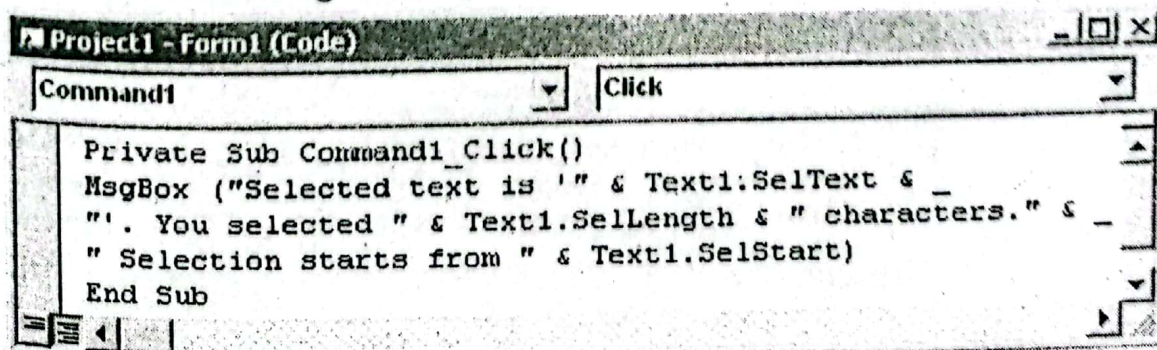
Figure: Change event example in execution**Program 8.3**

Write a program that displays a textbox and button on form. It displays selected text, length of selected text and starting position of selected text in a message box.

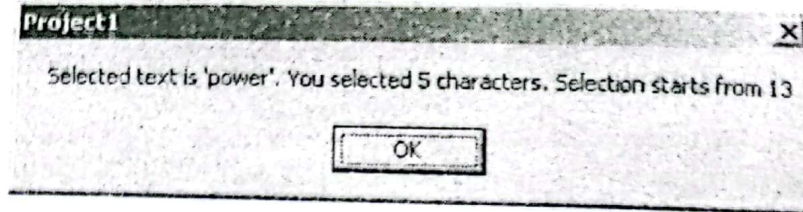
1. Start a new Standard EXE project.
2. Place a button and textbox on the form as follows:



3. Add the following code in the Click event of the button.



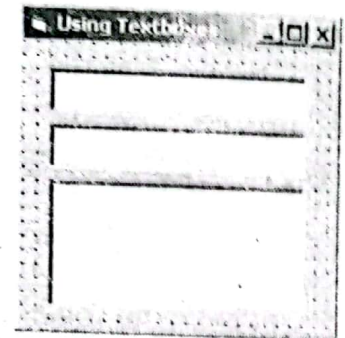
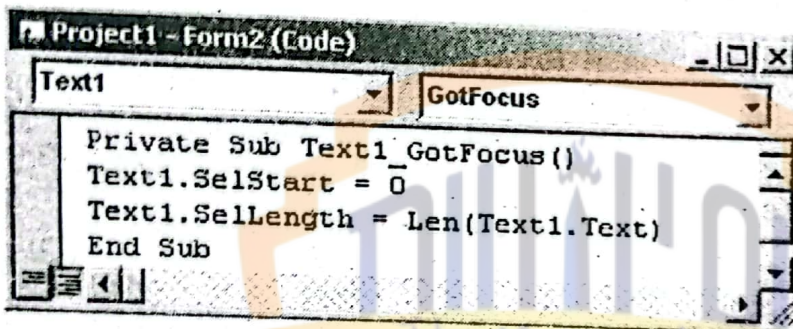
5. Select "power" from textbox and click button. The result will appear as follows:



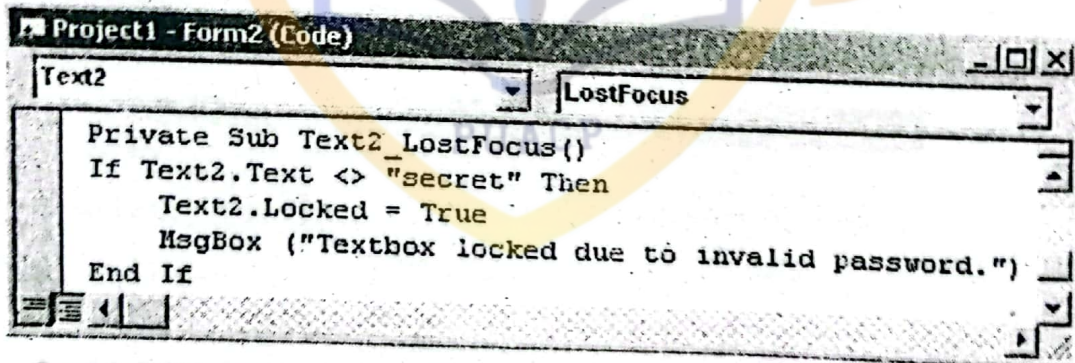
Program 8.4

Write a program that displays three textboxes to input name, password and comments. The contents of first textbox should be selected automatically when it gets focus. The second textbox should be locked if password is not "secret". It should be unlocked again if the user double clicks it. The third textbox should input comments in more than one line.

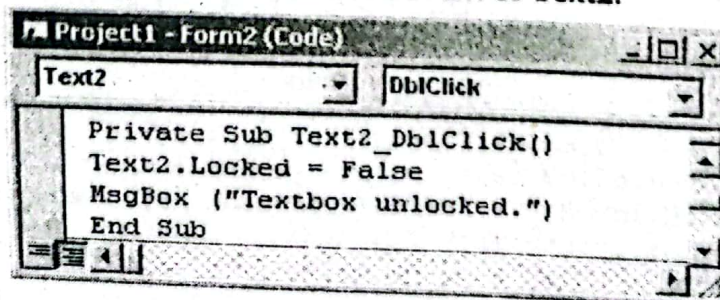
1. Start a new Standard EXE project.
2. Place three textboxes on the form.
3. Delete any text in Text property of all textboxes.
4. Add the following code in GotFocus event of Text1.



5. Set PasswordChar of Text2 to asterisk (*).
6. Add the following code in LostFocus event of Text2.



7. Set MultiLine property of Text3 to True.
8. Add the following code in DbClick event of Text2.

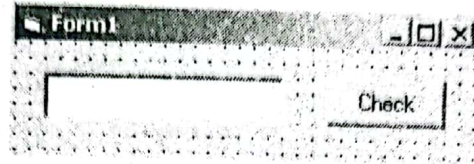


9. Run the project and test the working.

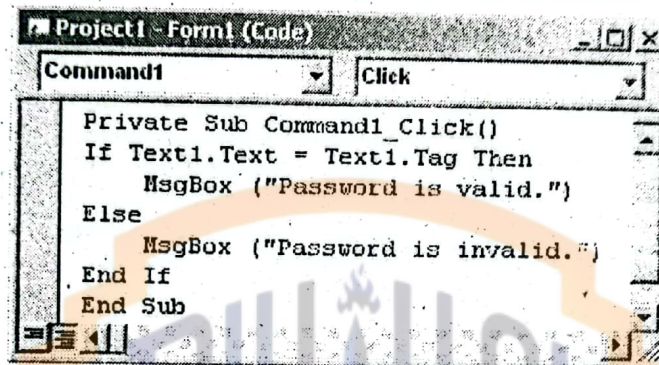
Program 8.5

Write a program that displays a textbox and a button. The user inputs password in the textbox. The program should validate the password by using Tag property of the textbox.

1. Start a new Standard EXE project.
2. Place a textbox and a button on the form.
3. Delete any text in Text property of the textbox.
4. Set Tag property of the textbox to "secret".
5. Set Caption of the button to "Check" as follows:



6. Add the following code in Click event of Command1.



7. Run the project and test the working.

8.5 Label Control

Label control is normally used to display messages to the user. For example, you can place a label with a textbox for describing the purpose of that textbox. The user cannot enter or edit the contents of a label. But you can change its contents from code window. You can also display the result of a calculation in a label control.

8.5.1 Important Properties of Label Control

Some important properties of label control are as follows:

AutoSize

This property is a Boolean property. If it is True, the size of the label control will change according to its contents. If it is False, its size will remain fixed.

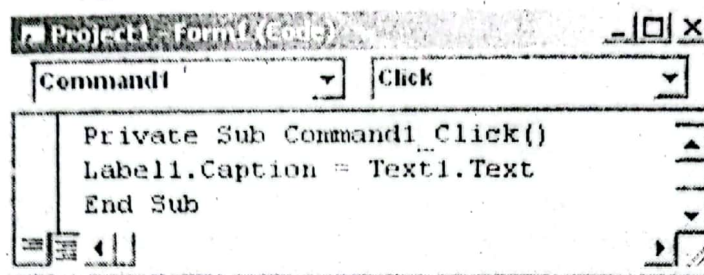
WordWrap

This property is a Boolean property. If it is True, the last word in the label will move to the next line if it does not fit in the current line. This property works with AutoSize property. If AutoSize is True and WordWrap is True, the last word will move to the next line and the label size will expand.

Program 8.6

Write a program that displays a label, a textbox and button on form. The contents of textbox should appear as caption of the label when the user clicks the button. The size of label should resize according to the length of the text. The user should not be able to enter more than 20 characters in the textbox.

1. Start a new Standard EXE project.
2. Place a button, textbox and label on the form.
3. Set **Caption** of the button to "Click Me".
4. Set **Text** property of textbox to "Hello World" and **MaxLength** to 20.
5. Set **Caption** property of label to "Hello" and **AutoSize** to **True**.
6. Add the following code in the **Click** event of the button.



7. Run the project. The form will appear as follows:



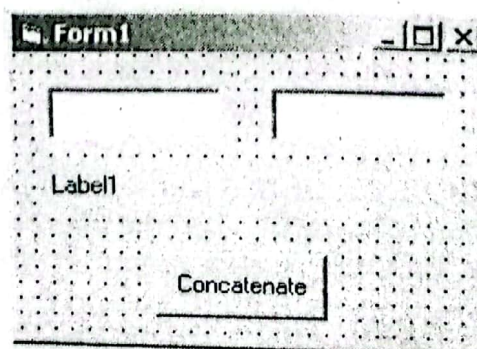
8. Click the button. The caption and size of label will change as follows:



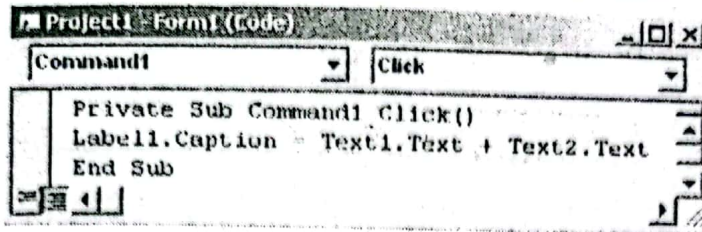
Program 8.7

Write a program that displays two textboxes, one label and button on form. The contents of two textboxes are concatenated and displayed on the label when the user clicks the button.

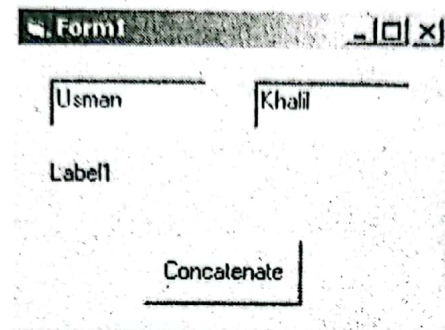
1. Start a new Standard EXE project.
2. Place a button, two textboxes and label on the form.
3. Set **Caption** of the button to "Concatenate".
4. Delete any text from **Text** property of both textboxes.
5. Set **AutoSize** property of label to **True**.
6. Design the form as follows:



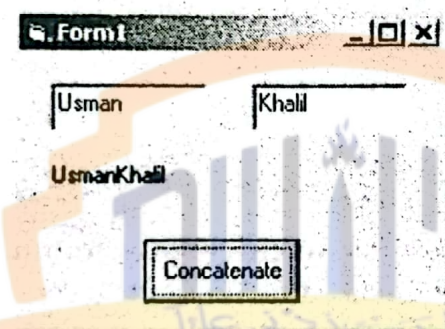
7. Add the following code in the Click event of the button.



8. Run the project and the following values:



9. Click the button. The caption of label will change as follows:



8.5.2 Important Events of Label Control

Some important events of Label control are as follows:

Click

The Click event occurs when the user clicks on the button.

DbClick

The DbClick event occurs when the user double clicks on the button.

Change

The Change event occurs when the contents of label control is changed.

8.6 Checkbox

Check Box control is used to give the user a choice of yes/no options. The Value property may be used to determine whether an item has been selected or not. The value 1 indicates that the checkbox is selected and the value 0 indicates that it is not selected. When the user selects a checkbox, ✓ sign appears in the box.

8.6.1 Important Properties of Checkbox

Some important properties of checkbox are as follows:

Caption

This property is used to display text with the checkbox.

Value

This property is used to determine whether the checkbox is selected or not. It may have three different values:

Value	Description
0	Checkbox is not selected
1	Checkbox is selected
2	Checkbox is disabled

Style

It is used to change the appearance of checkbox. It has two possible options Standard and Graphical.



Standard



Graphical

8.6.2 Important Events of Checkbox

Some important events of checkbox are as follows:

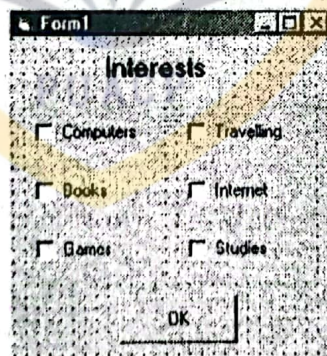
Click

The Click event occurs when the user clicks the checkbox or the checkbox gets focus.

Program 8 8

Write a program that uses different checkboxes to find out the interests of the user. It displays total number of interests checked by the user.

1. Start a new Standard EXE project.
2. Place a label on form and set its Caption to "Interests" and Font size to 14.
3. Place six checkboxes on form and design them as follows:



4. Add the following code in the Click event of the button.

```

Project1 - Form1 (Code)
Click
Command1
Private Sub Command1_Click()
    Dim n As Integer
    n = 0
    If Check1.Value = 1 Then n = n + 1
    If Check2.Value = 1 Then n = n + 1
    If Check3.Value = 1 Then n = n + 1
    If Check4.Value = 1 Then n = n + 1
    If Check5.Value = 1 Then n = n + 1
    If Check6.Value = 1 Then n = n + 1
    MsgBox "Your total interests are " & n
End Sub

```


5. Run the project.
6. Select the interests and click OK button. The result will appear as follows:

Program 8.9

Write a program that displays two textboxes and four checkboxes with captions "Addition", "Subtraction", "Multiplication" and "Division" respectively. The caption of checkboxes change when the user click them. For example, if the user selects "Addition" checkbox, the values of textboxes are added and result appear as the caption of the checkbox. The caption again changes to "Addition" if the user deselects it.

1. Start a new Standard EXE project.
2. Place two labels, two textboxes and four checkboxes.
3. Design the form as follows:

4. Add the following code in the Click event of Addition checkbox:

```

Private Sub chkAdd_Click()
    Dim a As Integer, b As Integer
    a = txtNo1.Text
    b = txtNo2.Text
    If chkAdd.Value = 0 Then
        chkAdd.Caption = "Addition"
    Else
        chkAdd.Caption = a + b
    End If
End Sub
  
```


5. Add the following code in the Click event of Subtraction checkbox:

```

Project1 - Form1 (Code)
chkSub Click
Private Sub chkSub_Click()
    Dim a As Integer, b As Integer
    a = txtNo1.Text
    b = txtNo2.Text
    If chkSub.Value = 0 Then
        chkSub.Caption = "Subtraction"
    Else
        chkSub.Caption = a - b
    End If
End Sub

```

6. Add the following code in the Click event of Multiplication checkbox:

```

Project1 - Form1 (Code)
chkMul Click
Private Sub chkMul_Click()
    Dim a As Integer, b As Integer
    a = txtNo1.Text
    b = txtNo2.Text
    If chkMul.Value = 0 Then
        chkMul.Caption = "Multiplication"
    Else
        chkMul.Caption = a * b
    End If
End Sub

```

7. Add the following code in the Click event of Division checkbox:

```

Project1 - Form1 (Code)
chkDiv Click
Private Sub chkDiv_Click()
    Dim a As Integer, b As Integer
    a = txtNo1.Text
    b = txtNo2.Text
    If chkDiv.Value = 0 Then
        chkDiv.Caption = "Division"
    Else
        chkDiv.Caption = a / b
    End If
End Sub

```

8. Run the project and enter values in textboxes as follows:

Form1

Enter No. 1

Enter No. 2

☐ Addition ☐ Multiplication

☐ Subtraction ☐ Division

9. Select different checkboxes. The result will appear as follows:

8.7 Option Buttons

Option buttons are used to give different choices to the user. The user can choose one of them. The user selects or deselects an Option button by clicking an Option button. It has a black dot inside when it is selected. Option button is also called radio button. The properties and events of option button are similar to checkbox.

Program 8.10

Write a program that uses two option buttons to get the gender of the user.

1. Start a new Standard EXE project.
2. Place a label on form and set its Caption to "Gender" and font size to 14.
3. Place two option buttons and set their Caption properties to "Male" and "Female" as follows:

4. Place a button on the form and set its Caption to "OK".
5. Add the following code in Click event of the button.

```

Project1 - Form1 (Code)
Click
Private Sub Command1_Click()
    If Option1.Value = True Then
        MsgBox "You are male"
    Else
        MsgBox "You are female"
    End If
End Sub

```

6. Run the project.
7. Select any option and click the button. The result will appear as follows:

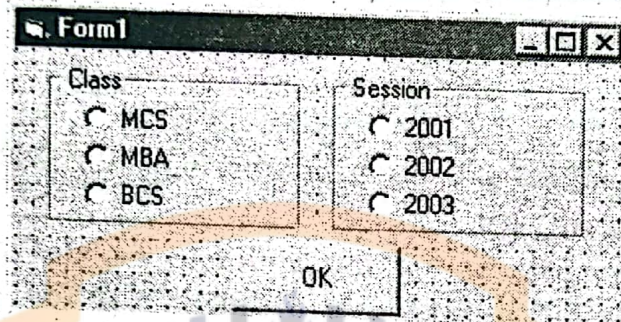
8.8 Frame

A frame is a rectangular region that holds other controls and groups them into a single set. For example you can place three option buttons in one frame and three in other frame. Now the user can select two options i.e. one from each frame. When one button is selected in a frame, other buttons are automatically deselected.

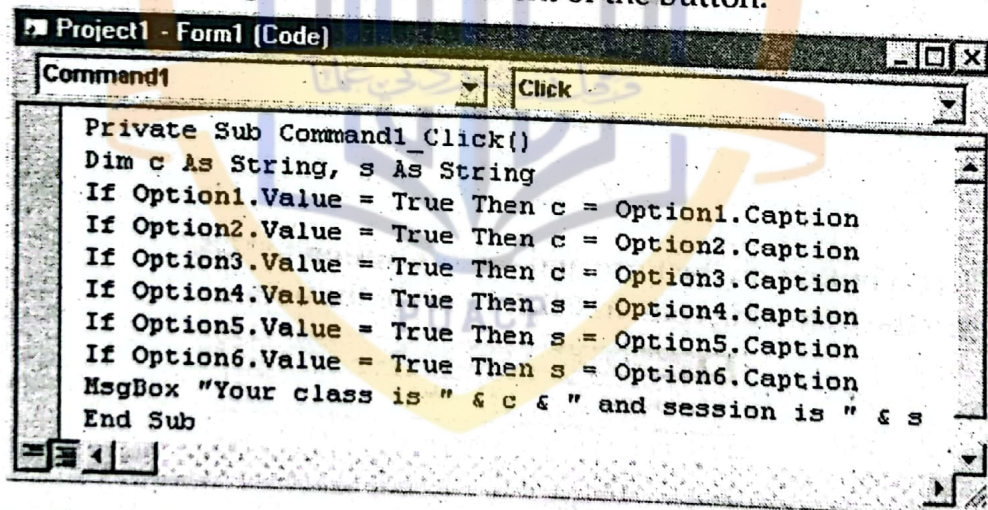
Program 8.11

Write a program that displays two frames with three options in each. One frame is used for class and the second for session. The user may selection one option from each frame.

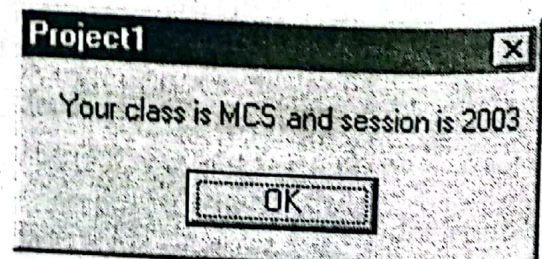
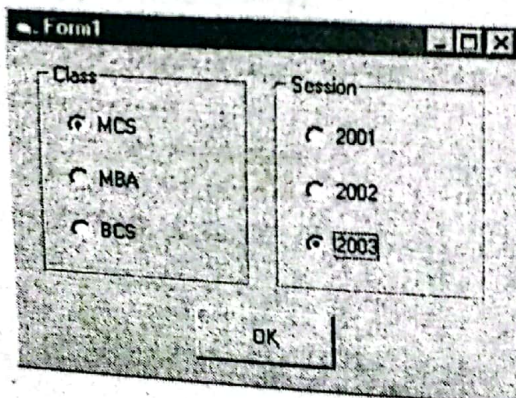
1. Start a new Standard EXE project.
2. Place two frames on the form.
3. Place a button and three option buttons in each frame as follows:



4. Set Value property of Option1 and Option4 to True in properties window.
5. Add the following code in Click event of the button.



6. Run the project.
7. Select any option from Class and Session frames and click OK. The result will appear as follows:



Program 8.12

Write a program that displays two frames and one label. First frame has seven option buttons with caption from 1 to 7. Second frame has three option buttons for colors Red, Green and Blue. The label displays name of week day selected from first frame in selected color from frame 2. For example if the user select 1 from first frame and Blue from second frame, the label displays "Friday" in blue color.

1. Start a new Standard EXE project.
2. Place two frames on the form.
3. Place two frames and one label on the form.
4. Place a control array of seven option buttons in first frame and a control array of three option buttons in second frame.
5. Set the properties of all controls as follows:

6. Add the following code in Click event of Option1 array of option buttons.

```

Project1 - Form3 (Code)
Option2 Click
Private Sub Option1_Click(Index As Integer)
    Select Case Index
    Case 0
        Label1.Caption = "Friday"
    Case 1
        Label1.Caption = "Saturday"
    Case 2
        Label1.Caption = "Sunday"
    Case 3
        Label1.Caption = "Monday"
    Case 4
        Label1.Caption = "Tuesday"
    Case 5
        Label1.Caption = "Wednesday"
    Case 6
        Label1.Caption = "Thursday"
    End Select
End Sub

```

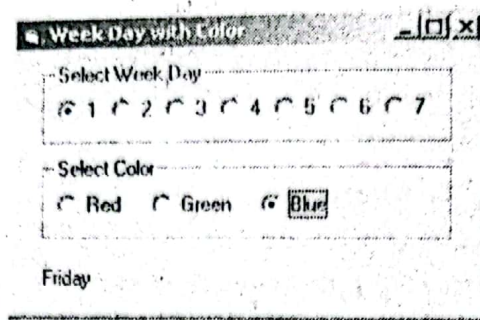
7. Add the following code in Click event of Option2 array of option buttons.

```

Project1 - Form3 (Code)
Option2 Click
Private Sub Option2_Click(Index As Integer)
    Select Case Index
    Case 0
        Label1.ForeColor = vbRed
    Case 1
        Label1.ForeColor = vbGreen
    Case 2
        Label1.ForeColor = vbBlue
    End Select
End Sub

```


8. Run the project.
9. Select 1 from first frame and Blue from second frame. The result will appear as follows:



8.9 List Box Control

ListBox displays a list of items that may be selected by the user. If the number of items in the list exceeds the size of the list box, scroll bars are added automatically. The user does not have to type text in textbox. It also restricts the user to select only from available options.

8.9.1 Important Properties of List Boxes

Some important properties of list box control are as follows:

Columns

This property determines the number of columns in the list box. If 0, the ListBox scrolls vertically in a single column. If 1 or more, the ListBox items appear in the number of columns specified (one or more columns) and a horizontal scrollbar appears to see all the items in list.

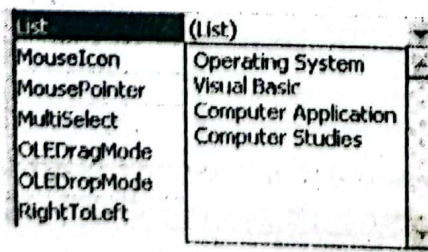
Index

This property is used to determine the index of the item that is most recently added in the list box. This property is available only at run time.

List

This property can be used both in design time and run time. At design time, you can enter different values in the list box by using the following procedure:

1. Select the list box control.
2. Click the List property in the properties windows.
3. Type any value and press **CTRL+ENTER** to enter another item.
4. Press **ENTER** when finished.



At run time, List is used as an array containing all items of the list box. You can display any item of the list box by using this property. The following code will display the second item of the list box.

```
MsgBox List1.List(1).
```

ListCount

This property is used to determine the total number of items in a list box. This property is available only at run time.

ListIndex

This property is used to determine the most recently selected item in the list. The value of -1 indicates that no item is selected yet. This property is available only at run time.

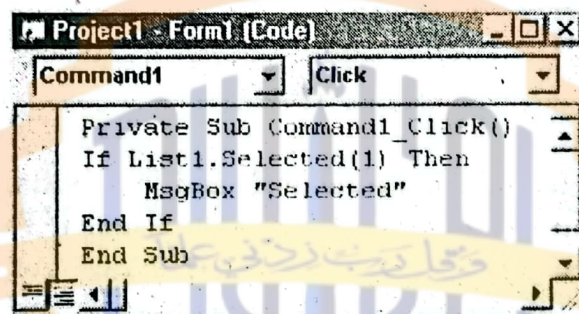
MultiSelect

This property is used to specify the selection rules of the list box.

Value	Description
0-None	The user can select only one item by clicking with mouse or pressing SPACE
1-Simple	The user can select multiple items by clicking with mouse or pressing SPACE
2-Extended	The user can select multiple items by holding SHIFT key and clicking on different items.

Selected

This property is used to determine whether a particular item of the list box is selected or not. This property is used as an array with index. This property is available only at run time. The following line will check whether the second item is selected or not.



Sorted

This is Boolean property. If **True**, it sorts the values of a ListBox automatically. If **False** (default value), the values appear in the same order in which the program added the items to the list.

SelCount

This property indicates the number of selected items in the list. It is useful only if the **MultiSelect** property is used.

Style

This property is used to display the list box in different styles.

Value	Description
0-vblistboxStandard	The list box is displayed in default setting
1-vblistboxCheckbox	The list box contains a checkbox on left side of each item

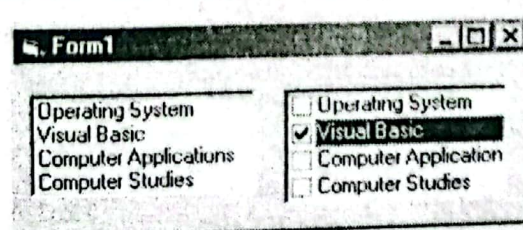


Figure: Two styles of list box

Text

This property is used to determine the value of the most recently selected item in the list box. This property is available only at run time.

8.9.2 Important Methods of List Box

Some important methods of list box are as follows:

AddItem

This method is used to add an item in the list box at run time. For example, the following code will add "IT Series" in the list box.

```
List1.AddItem "IT Series"
```

Clear

This method is used to remove all items from the list box.

```
List1.Clear
```

RemoveItem

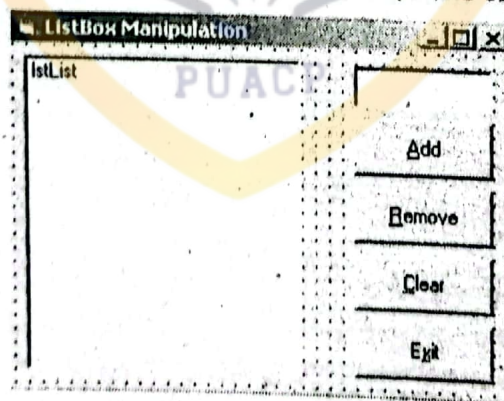
This method is used to remove a particular item from the list box. This method accepts the index of the item as parameter. For example the following line will remove the second item of the list box.

```
List1.RemoveItem 1
```

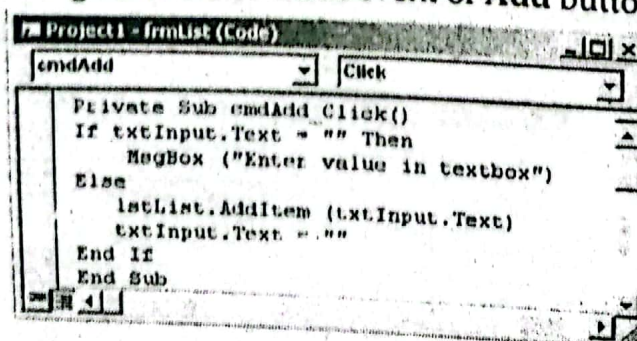
Program 8.13

Write a program that uses a list box, textbox and four buttons. The user can enter a text in the textbox and press a button to add the text in the list box. The second is used to remove an item from the list. The third button is used to clear the list box and the last button is used to exit the program.

1. Start a new Standard EXE project.
2. Place a list box, a textbox and four buttons on the form as follows.



3. Add the following code in the Click event of Add button.



4. Add the following code in the Click event of Remove button.

```
Project1 - frmList (Code)
cmdRemove Click
Private Sub cmdRemove_Click()
    If lstList.ListIndex <> -1 Then
        lstList.RemoveItem (lstList.ListIndex)
    End If
End Sub
```

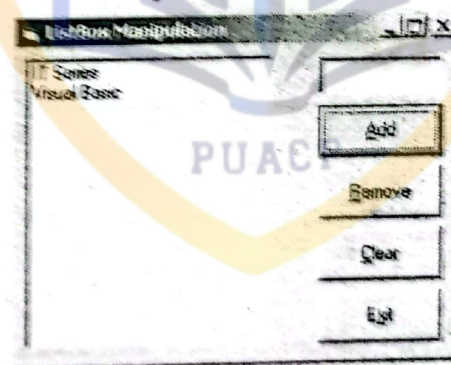
5. Add the following code in the Click event of Clear button.

```
Project1 - frmList (Code)
cmdClear Click
Private Sub cmdClear_Click()
    lstList.Clear
End Sub
```

6. Add the following code in the Click event of Exit button.

```
Project1 - frmList (Code)
cmdExit Click
Private Sub cmdExit_Click()
    End
End Sub
```

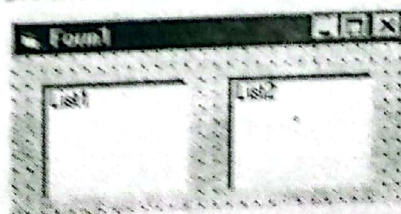
7. Run the project.
8. Add any item in the text box and click Add button. The text will be added in the list box. You can add as many items as needed.



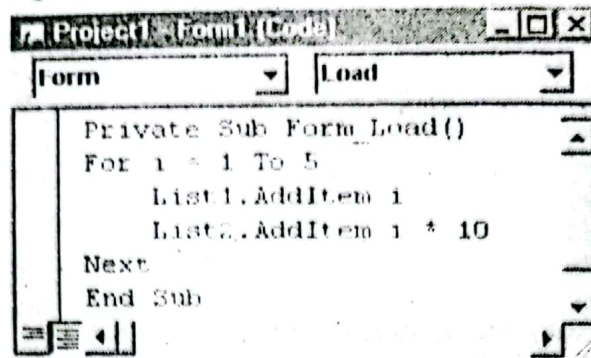
Program 8.14

Write a program that uses two list boxes. Each list box is filled automatically with five items when the project starts. The user can move an item of one list box to other by double clicking that item.

1. Start a new Standard EXE project.
2. Place two list boxes on the form.



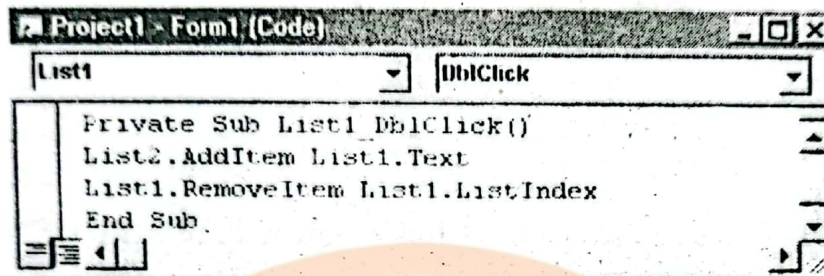
3. Add the following code in the Load event of the form.



```

Project1 - Form1 (Code)
Form Load
Private Sub Form_Load()
    For i = 1 To 5
        List1.AddItem i
        List2.AddItem i * 10
    Next
End Sub
  
```

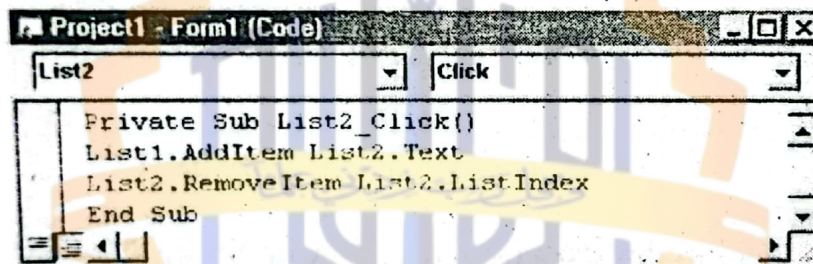
4. Add the following code in the DblClick event of List1.



```

Project1 - Form1 (Code)
List1 DblClick
Private Sub List1_DblClick()
    List2.AddItem List1.Text
    List1.RemoveItem List1.ListIndex
End Sub
  
```

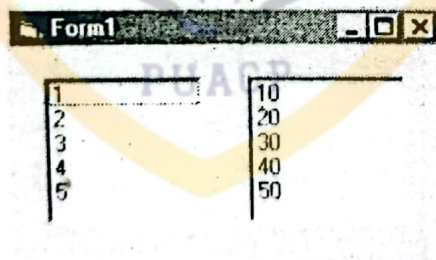
5. Add the following code in DblClick event of the List2.



```

Project1 - Form1 (Code)
List2 Click
Private Sub List2_Click()
    List1.AddItem List2.Text
    List2.RemoveItem List2.ListIndex
End Sub
  
```

6. Run the project. The form will appear as follows:



1	10
2	20
3	30
4	40
5	50

7. Double any item of any list box. It will move to the other list box.

How It Works?

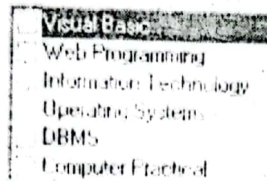
When the above example is executed, the code in the Load event is executed. It fills both list boxes with different values. When the user double clicks any item, Text method determines the text of the item and adds in the other list box. ListIndex method is used to remove the item from the double clicked list.

Program 8.15

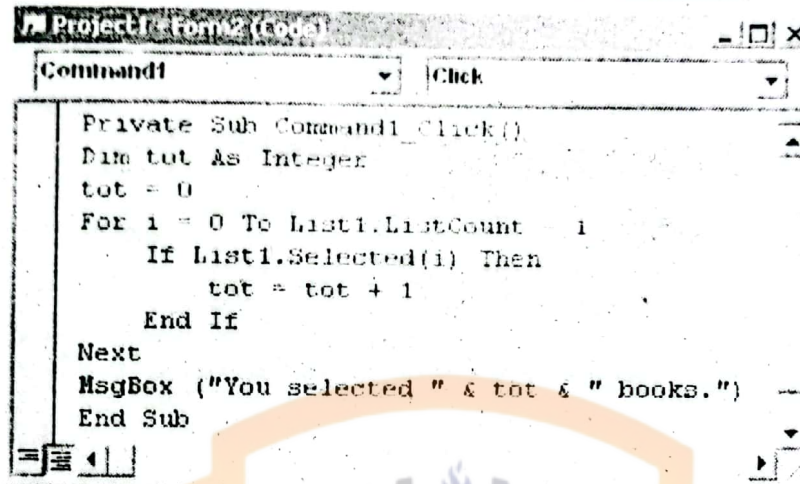
Write a program that displays a list box of checkbox style and a button. The program displays total number of selected items when the user clicks the button.

1. Start a new Standard EXE project.
2. Place a list box and a button on the form.

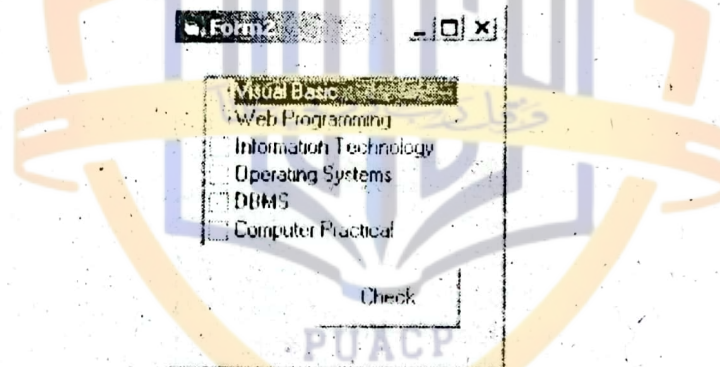
3. Set Caption of the button to "Check".
4. Set Style of List1 to "1 - Checkbox".
5. Add the following values in the list using List property.



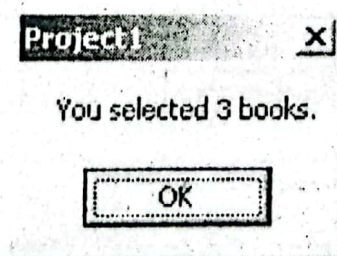
6. Add the following code in the Click event of Command1.



7. Run the project. The form will appear as follows:



8. Select Visual Basic, Web Programming, DBMS and click the button. The result will appear as follows:



8.10 Combo Box Control

A combo box is a collection of list box and textbox. Combo box is similar to a list box. List, ListIndex, ListCount, Index, and other properties work the same way, however ComboBoxes cannot have multiple columns or handle multiple selections.

The user can select an item from combo box and type in textbox. Visual Basic provides three types of combo boxes. It is specified by using **Style** property in the properties window.

1. Drop Down Combo Box

This type of combo box displays only one item unless the user clicks the ▼ button to display additional items. The user can also enter values at the top of the Combo Box in the same way you do in a Text Box.

2. Simple Combo Box

This type of combo box looks like a ListBox attached to a TextBox. Items are displayed as if they were in a ListBox. You may also enter values on top of the ComboBox.

3. Drop Down List Box

This type of combo box does not allow the user to enter values. It is similar to ListBox.

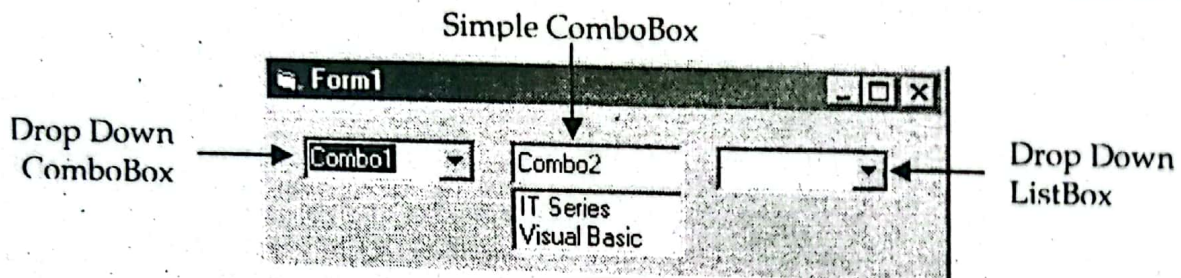
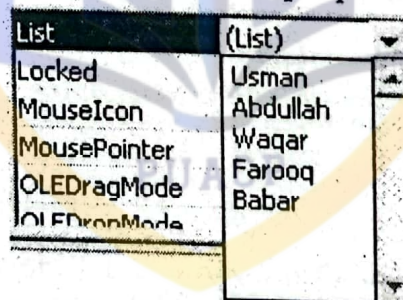


Figure: Three styles of combo box

Program 8.16

Write a program that a combo box to display five string values and a button with caption "Search". It get a value from the user and searches it from the combo box.

1. Start a new Standard EXE project.
2. Place a combo box and a button on the form.
3. Add five values in the combo box from properties window as follows:



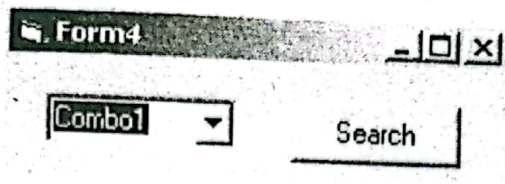
4. Set Caption of Command1 to "Search".
5. Add the following code in the Click event of Command1.

```

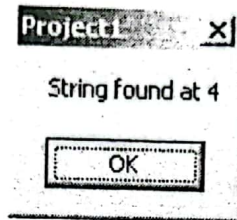
Project1 - Form4 (Code)
Command1 Click
Private Sub Command1_Click()
    Dim n As String
    n = InputBox("Enter a string:", "Search", "Usman")
    For i = 0 To Combo1.ListCount - 1
        If n = Combo1.List(i) Then
            MsgBox ("String found at " & i)
            Exit Sub
        End If
    Next
    MsgBox ("String not found.")
End Sub

```


6. Run the project. The form will appear as follows:



7. Click **Search** button. The input box will appear to get a value.
 8. Enter "Babar" and click OK. The result will appear as follows:



Note: The same procedure can be used to search a listbox.

8.11 Image and Picture Controls

Image and Picture controls are used to display different types of pictures on the form. Image control takes less memory than picture control. Different properties of image and picture controls are similar.

8.11.1 Important Properties of Picture Control

Some important properties of Picture control are as follows:

Picture

Picture property is used to select the graphic to be displayed in the picture control.

AutoSize

It is a **Boolean** property. If it is **False**, the control does not resize to fit the image. If it is **True**, the control resizes and does not truncate the image. So the PictureBox's size depends on the size of the image to be displayed.

8.11.2 Important Properties of Image Control

Some important properties of Image control are as follows:

Picture

Picture property is used to select the graphic to be displayed in the image control.

Stretch

The ImageBox has a unique Boolean property called **Stretch**. If it is **True**, the graphic resizes to fit the size of the ImageBox.

8.11.3 Loading Pictures at Run Time

You can load pictures in image or picture controls in two ways:

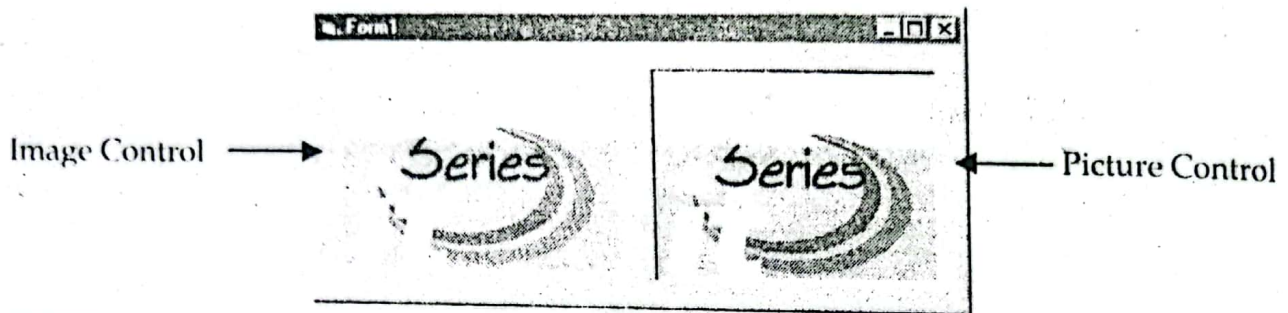
- Selecting an image by using the Picture property in the properties window.
- Using the LoadPicture method at run time. The syntax of this method is:

```
Picture1.Picture = LoadPicture("C:\MyPic.jpg")
```

Program 8.17

Write a program that displays the same picture in image and picture controls.

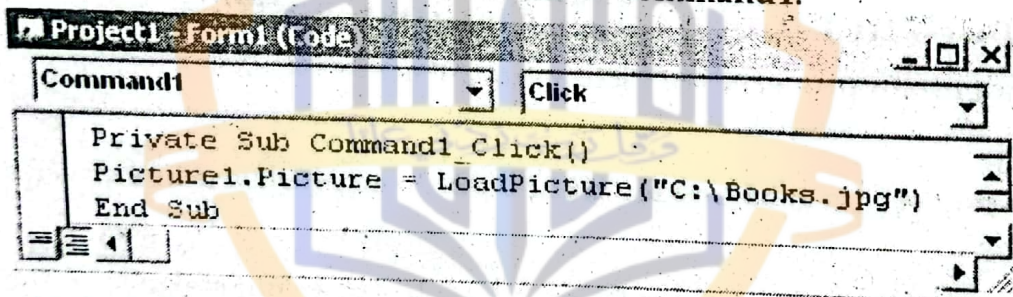
1. Start a new Standard EXE project.
2. Place an image control and a picture control on the form.
3. Select any picture by using **Picture** property in properties window.
4. Run the project. The picture will appear as follows:



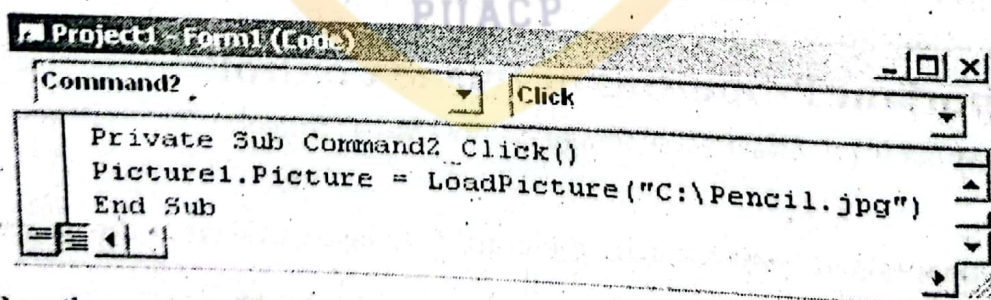
Program 8.18

Write a program that displays two buttons and one picture box. The contents of picture box changes with the click of buttons.

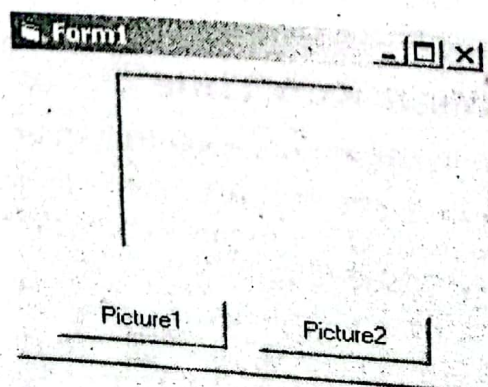
1. Start a new Standard EXE project.
2. Place a picture box and two buttons on the form.
3. Set Caption of **Command1** to "Picture1" and **Command2** to "Picture2"
4. Set **AutoSize** of picture box to **True**.
5. Add the following code in Click event of **Command1**:



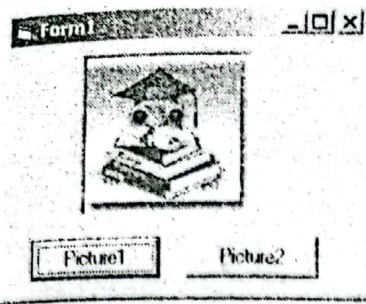
6. Add the following code in Click event of **Command2**:



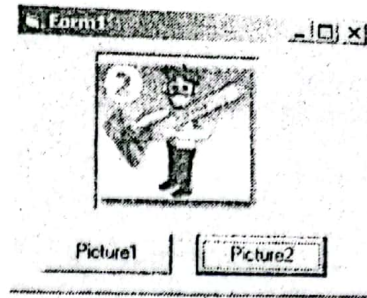
7. Run the project. The form will appear as follows:



8. Clicking on buttons will display different picture in the picture box as follows:



Picture1 button clicked



Picture2 button clicked

8.12 Scrollbar Control

Scrollbar control is used to move vertically or horizontally. Visual Basic provides two types of scrollbars.

- Vertical Scrollbar
- Horizontal Scrollbar



Figure: Horizontal Scrollbar

8.12.1 Important Properties of Scrollbar

Some important properties of scrollbar control are as follows:

LargeChange

This property specifies the amount of change in **Value** property of scrollbar when the user clicks within the scrollbar's shaft area.

Max

This property indicates the maximum value of scrollbar **Value** property. The default value is 32767.

Min

This property indicates the minimum value of scrollbar **Value** property. The default value is 0.

SmallChange

This property specifies the amount of change in **Value** property of scrollbar when the user clicks an arrow at either end of the scrollbar.

Value

This property contains the current position of the scrollbar.

8.12.2 Important Events of Scrollbar

Some important events of scrollbar control are as follows:

Change

This event occurs when the position of scrollbar is changed.

Scroll

This even occurs when the scrollbar is being scrolled.

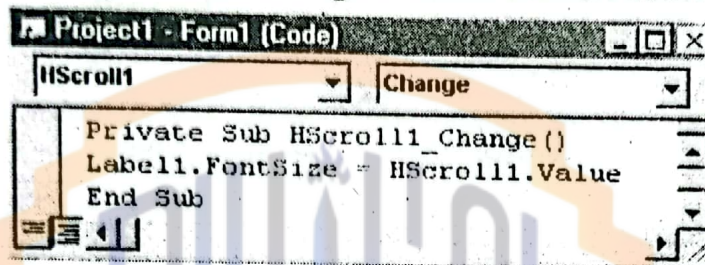
Program 8.19

Write a program that uses a scrollbar and a label. The user can change the size of the label by clicking on the scrollbar.

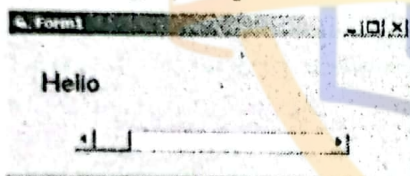
1. Start a new Standard EXE project.
2. Place a textbox on form and set its **Caption** to "Hello" and font size to 14.
3. Place a scrollbar on the form and set its **Min** value to 14, **Max** value to 36, **SmallChange** to 1 and **LargeChange** to 4 as follows:



4. Add the following code in **Change** event of the scrollbar.



5. Run the project.
6. Change the position of the scrollbar. The size of the label will also change.



Size of label before change

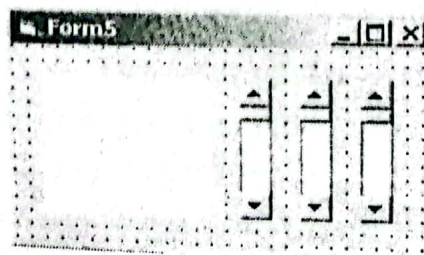


Size of label after change

Program 8.20

Write a program that uses three scrollbars and label. The scrollbars are used to specify the amount of colors. The first scrollbar refers to Red, second refers to Green and third refers to Blue color. The user can change the color of label through scrollbars using RGB function.

1. Start a new Standard EXE project.
2. Place a label on form and delete any text from its **Caption** property.
3. Place three vertical scrollbars on the form.
4. Set **Min** value to 0, **Max** value to 255, **SmallChange** to 1 and **LargeChange** to 25 of all scrollbars:



5. Add the following code in Change event of all scrollbars:

```
Label1.BackColor = RGB(VScroll1.Value, VScroll2.Value, VScroll3.Value)
```

6. Run the project.

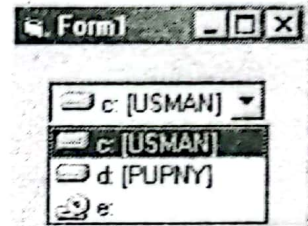
7. Change the position of scrollbars. The back color of the label will also change.

How it Works?

The above Example uses RGB function to set the color of label. The RGB is a built-in function that accepts three parameters for Red, Green and Blue colors and creates a color of these three combinations.

8.13 DriveListBox Control

DriveListBox control is used to display all drives on a computer system. A list of all available drive of computer is displayed in a list box. The user can select any drive from the list box. When you place a drive list box on a form, it automatically access all drives present on the computer. You do not need to write any code for this purpose.



8.13.1 Important Properties of DriveListBox

Important properties of DriveListBox are as follows:

Drive

This property is used to determine the name of the selected drive.

8.13.2 Important Events of DriveListBox

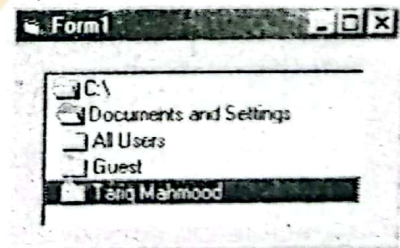
Important events of DriveListBox are as follows:

Change

This event occurs when the user changes the selected drive.

8.14 DirListBox Control

DirListBox is used to display a list of all directories on a drive. When you place this control on form, it automatically displays the directories of computer drive. You do not need to write any code for this purpose. You can open any directory by double clicking on it.



8.14.1 Important Properties of DirListBox

Important properties of DirListBox are as follows:

Path

This property is used to determine the path of selected directory.

8.14.2 Important Events of DirListBox

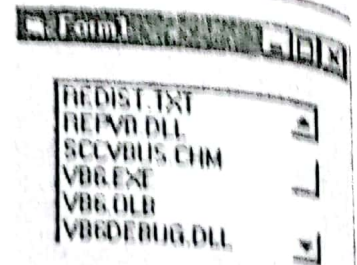
Important events of DirListBox are as follows:

Change

This event occurs when the user changes the selected directory.

8.15 FileListBox Control

FileListBox is used to display all files in current directory. When this control is used on form, it automatically displays all files in a list without writing any code.



8.15.1 Important Properties of FileListBox

Some important properties of FileListBox are as follows:

FileName

This property is used to determine the name of the selected file.

Archive

It is a Boolean property. It specifies whether archive file is displayed or not. By default, it is **True**.

Hidden

It is a Boolean property. It specifies whether hidden file is displayed or not. By default, it is **False**.

MultiSelect

This property specifies whether the user can select multiple files or not.

Path

This property is used to determine the path of the selected file.

Pattern

This property is used to filter the files to be displayed in the FileListBox.

ReadOnly

This property specifies whether read-only file is displayed or not. By default, it is **True**.

System

This property specifies whether system file is displayed or not. By default, it is **True**.

8.15.2 Important Events of FileListBox

Some important events of FileListBox are as follows:

Click

This event occurs when the user clicks FileListBox.

DblClick

This event occurs when the user double clicks FileListBox.

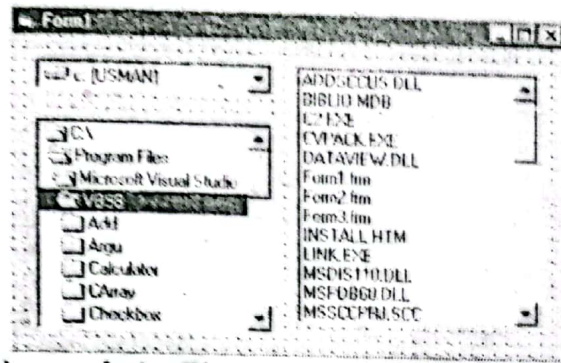
Combining DriveListBox, DirListBox & FileListBox

These three controls are used together to access files in a computer system. All drives are displayed in DriveListBox. All directories in that drive appear in DirListBox when a drive is selected. All files in that directory appear in FileListBox when a directory is selected.

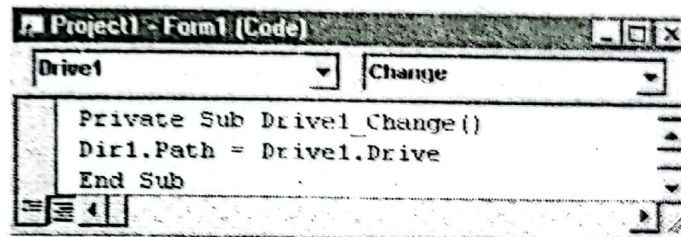
Program 8.21

Write a program that uses DriveListBox, DirListBox, and FileListBox to provide access to any file on computer.

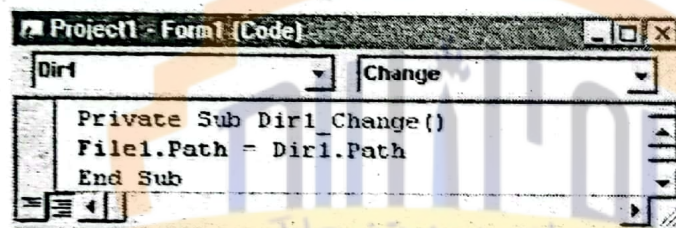
1. Start new Standard EXE project.
2. Place a DriveListBox, a DirListBox and a FileListBox on form as follows:



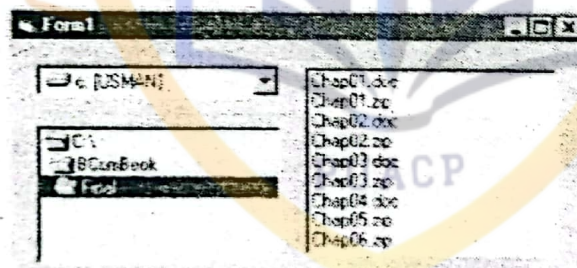
3. Add the following code in Change event of Drive1.



4. Add the following code in Change event of Dir1.



5. Run the project and select any drive and directory. It will work as follows:



8.16 Timer Control

Timer control is used to execute some statements after a certain time period. For example, it can be used to change the color of Form after every 10 seconds or display a message after every 5 minutes etc. It works in background and is not visible at run time.

8.16.1 Important Properties of Timer

Some important properties of Timer are as follows:

Interval

This property specifies the number of milliseconds after which the event procedure of the timer will be executed. It must be between 1 and 65535.

Enabled

It is a Boolean property. If it is true, the code written in Timer event continues to execute after the specified time. If it false, the working of timer is disabled.

8.16.2 Important Events of Timer

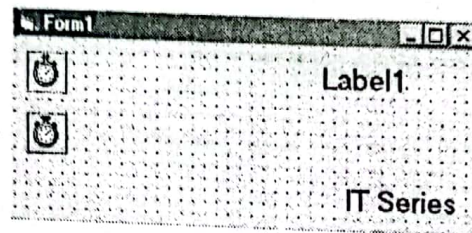
Timer

The Timer event occurs after specified milliseconds in Interval property.

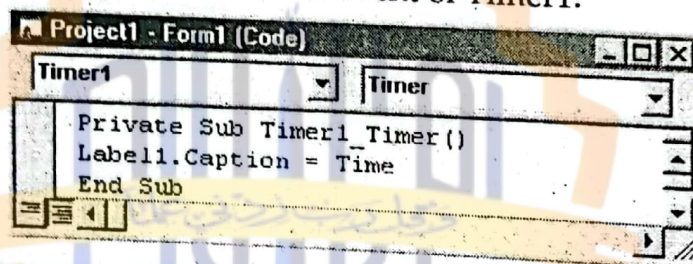
Program 8.22

Write a program that uses two timers. One time is used to display the current time. The second time is used to a scrolling message on the form.

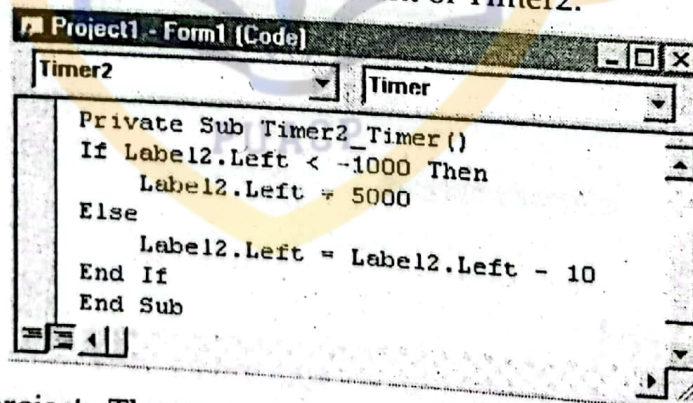
1. Start a new Standard EXE project.
2. Place two Labels and two Timers on the form.
3. Set the font size of both labels to 14 and Caption of Label2 "IT Series".



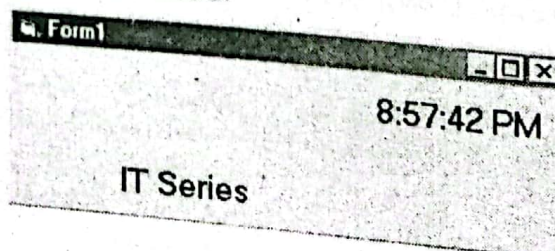
4. Set the Interval property of Timer1 and Timer2 to 1.
5. Add the following code in Timer event of Timer1.



6. Add the following code in Timer event of Timer2.



7. Run the project. The current time will appear in Label1. The Label2 will be scrolling from right to left side.

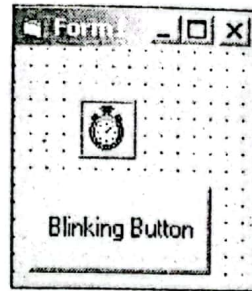


Program 8.23

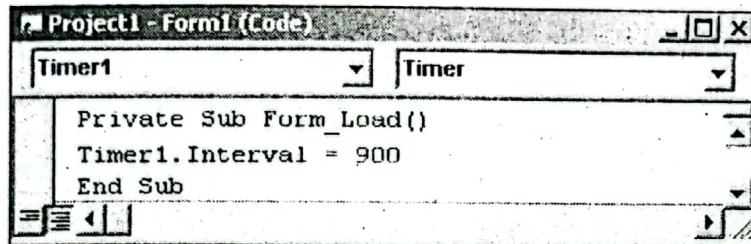
Write a program that applies blinking effect to a command button.

1. Start a new Standard EXE project.

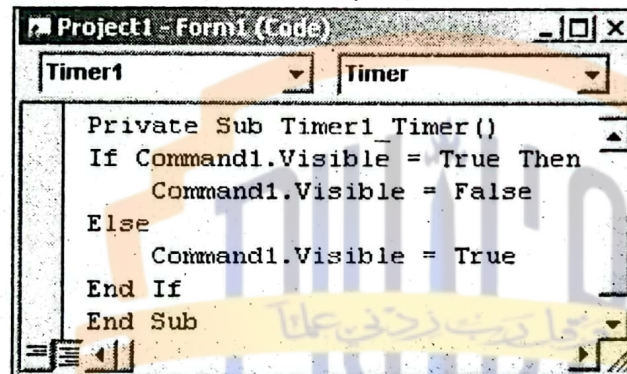
2. Design the form as follows:



3. Double click the form and add following code in its Load event.



4. Double click the timer and add following code in its Timer event.



5. Run the project and view the blinking button.

8.17 Shape Control

Shape control is used to display different types of shapes like square, rectangle circle.

8.17.1 Important Properties of Shape

Some important properties of Shape control are as follows:

BackStyle

This property is used to determine whether the background of the shape is transparent or opaque (solid).

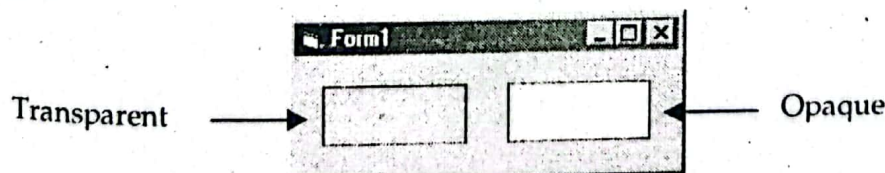


Figure: Two options of BackStyle

FillStyle

This property is used to determine the fill style of the shape.

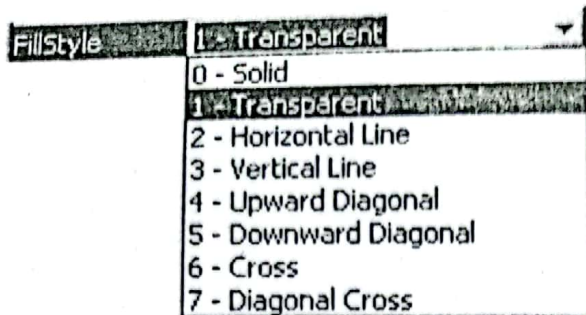


Figure: FillStyle options

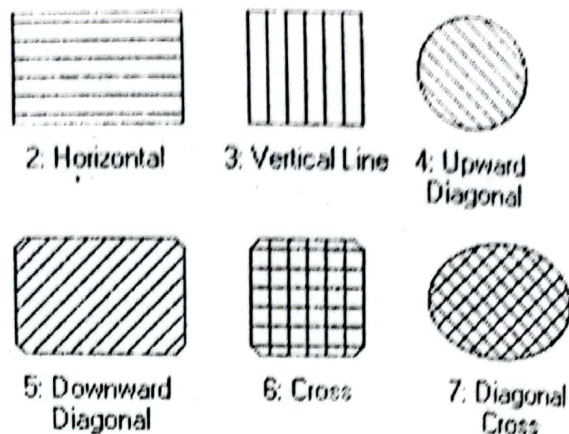


Figure: Fill Styles

Shape

This property is used to determine the type of shape.

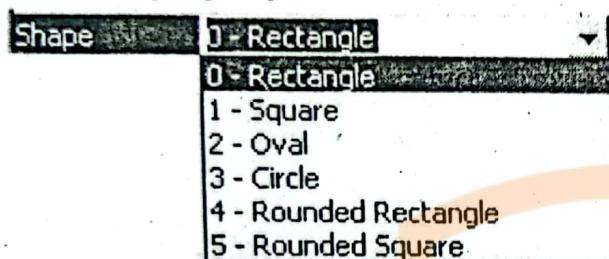


Figure: Shape options

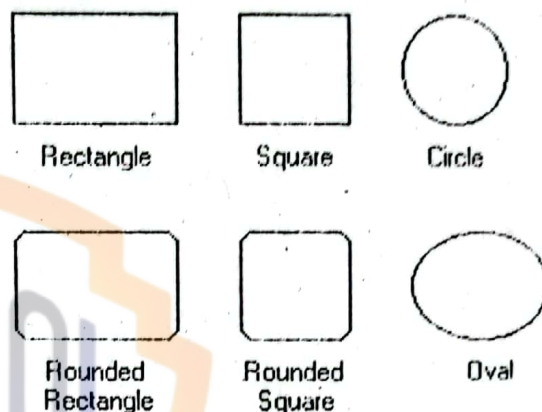
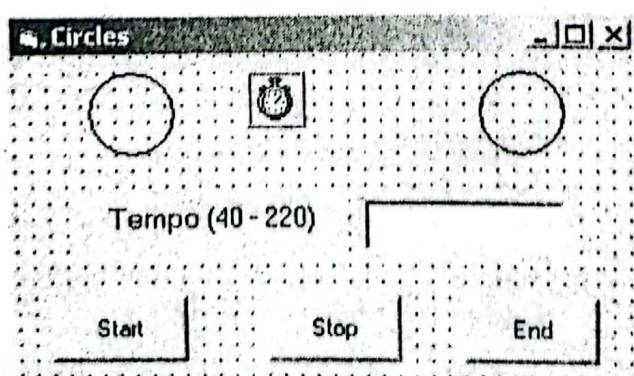


Figure: Shapes

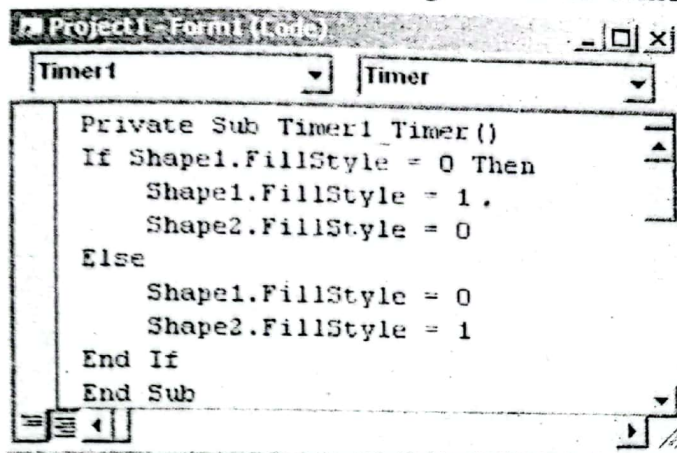
Program 8.24

Write a program that displays two circles, one textbox and three buttons. It gets speed from user in textbox. It fills circles one by one with given speed as the user click Start button. It stops doing this as the user clicks Stop button. The program terminates when the user clicks End button.

1. Start a new Standard EXE project.
2. Place two circles, one timer, one label, one textbox and three buttons on the form.
3. Design the form as follows:



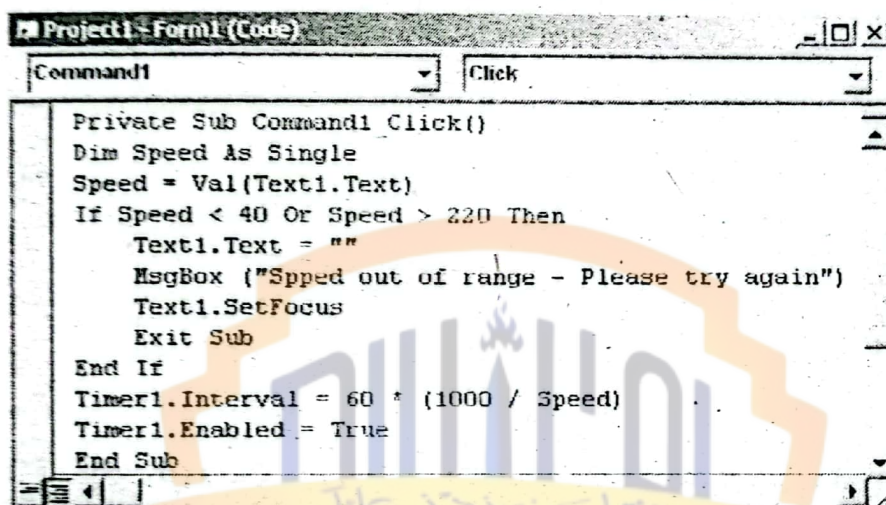
4. Double click the timer and add following code in its Timer event.



```

Project1 - Form1 (Code)
Timer1 Timer
Private Sub Timer1_Timer()
    If Shape1.FillStyle = 0 Then
        Shape1.FillStyle = 1
        Shape2.FillStyle = 0
    Else
        Shape1.FillStyle = 0
        Shape2.FillStyle = 1
    End If
End Sub
  
```

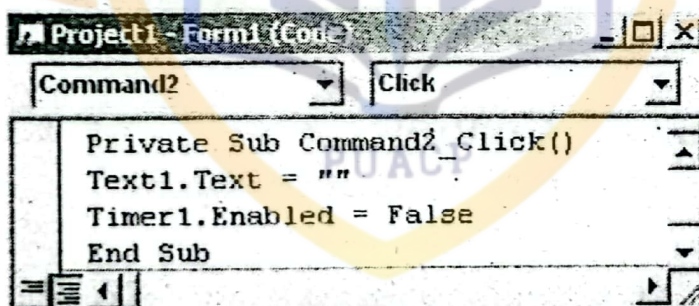
5. Add the following code in Click event of Start button.



```

Project1 - Form1 (Code)
Command1 Click
Private Sub Command1_Click()
    Dim Speed As Single
    Speed = Val(Text1.Text)
    If Speed < 40 Or Speed > 220 Then
        Text1.Text = ""
        MsgBox ("Speed out of range - Please try again")
        Text1.SetFocus
        Exit Sub
    End If
    Timer1.Interval = 60 * (1000 / Speed)
    Timer1.Enabled = True
End Sub
  
```

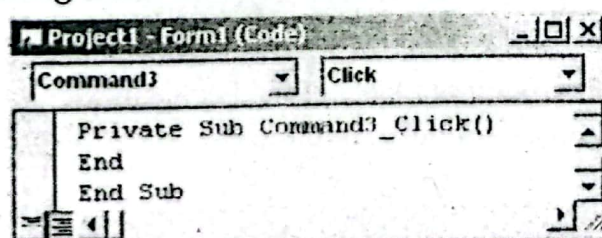
6. Add the following code in Click event of Stop button.



```

Project1 - Form1 (Code)
Command2 Click
Private Sub Command2_Click()
    Text1.Text = ""
    Timer1.Enabled = False
End Sub
  
```

7. Add the following code in Click event of End button.



```

Project1 - Form1 (Code)
Command3 Click
Private Sub Command3_Click()
End Sub
  
```

8. Run the project and enter 100 in textbox.
9. Click Start button and check the working.

8.18 Line Control

Line control is used to draw different types of lines on the form. You can change the width of the line by using **BorderWidth** property.

8.19 OLE Control

OLE stands for Object Linking & Embedding. OLE control is used to avail the facilities of an application in a project by embedding it in OLE object. In this situation, project is called **Container Application** and embedded application is called **Server Application**.

8.19.1 Methods of OLE Control

Close

This method is used to close the active server application.

Syntax: OLE.Close()

CreateEmbed

It creates an embedded object from specified file and associates it with OLE control. It enables an existing file to be embedded in the application.

OLE1.CreateEmbed ("c:\Test.doc")

This above line associates Word object created from file **Test.doc** the control OLE1.

CreateLink

It creates and link object from the specified file and associates it with OLE control. It also enables an existing file to be embedded in the application.

OLE1.CreateLink("c:\Test.doc")

This above line links the file **Test.doc** with the control OLE1.

InsertObjDlg

It displays **Insert Object** dialog box to select a type of object to create. When user clicks OK, an object of selected type is created and associated with OLE control. It enables the user to use any registered server with application. The function reads the registry for a list of registered servers each time it is displayed.

OLE1.InsertObject()

8.19.2 Properties of OLE Control

Some important properties of OLE control are as follows:

AutoActivate

This property decides how to activate the server. The options are on **double-click**, **get focus**, **automatic** and **manual**.

Class

It is used to set or get the name of object class. For example, if the user associates Word document with control OLE1, class property will have a value: 'Word.Document'.

Object

It returns a reference to OLE object held in OLE control.

8.19.3 Events of OLE Control

Some important events of OLE control are as follows:

ObjectMove

This event is called whenever the active object is moved.

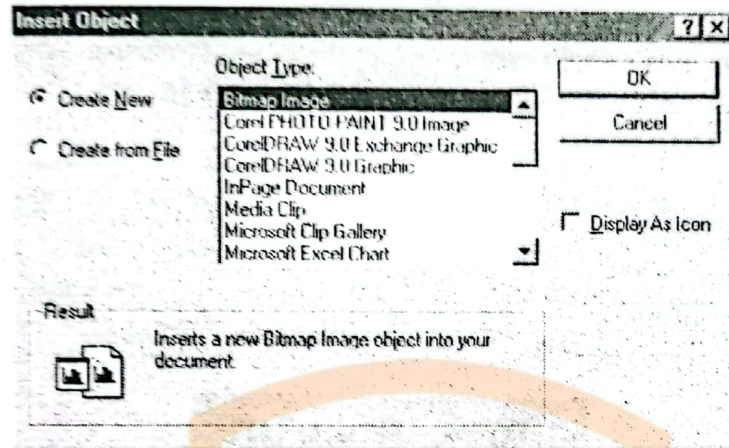
Update

This event occurs when the object's data is changed. It occurs every time the server object is active and the user changes it in any manner.

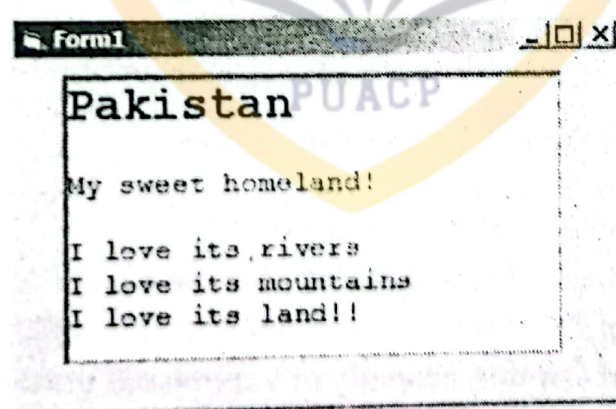
Program 8.25

Write a program that uses an OLE object to embed WordPad application in project.

1. Start a new Standard EXE project.
2. Place an OLE object on the form. An Insert Object dialog box will appear.



3. Click **Cancel** button to insert an empty OLE object on the form.
4. Start WordPad and open any file such that both WordPad and Visual Basic Form are visible.
5. Select all contents of the file and drag it on the OLE object in Visual Basic. The file will be inserted in the OLE object.
6. Run the project and double click the OLE object. The menu bar WordPad will appear on the form. You can use menu bar to format the file in OLE object.



Programming Exercise

1. Write a program that contains a textbox, a list box and a button. When the user clicks the button after entering some text in the textbox, the value of the textbox should be added in the list box and total number of items in the list should appear on the title bar of the form.
2. Write a program that contains four option buttons, two textboxes and a button. Each option button represents an arithmetic operator. The user enters two numbers in the textboxes, selection an option button for the desired operation and clicks the button. The program should apply the same operation as selected by the user.
3. Write a program that uses a combo box to display five cities of Pakistan. When the user clicks on a city in the combo box, the name of the city should appear in a message box.

Important Note: The solutions of all questions are available at the end of the book.

Short Questions

Q.1. What are Intrinsic Controls?

The controls which are automatically present in the toolbox when you start Visual Basic project are called intrinsic controls. The intrinsic controls are available to every VB program you write.

Q.2. What do you mean by default property of a control? Give an example.

Each control has one property that is the default property. When the default property of a control is used, it is not necessary to write the name of the property. For example, Text property is the default property for text box. The following two statements are equivalent:

Text1.Text = "IT Series"

Text1 = "IT Series"

Q.3. What is the purpose of "Name" property of a control?

Name property is used to refer to the control in Visual Basic code. The name should be meaningful. It should indicate the purpose of the control.

Q.4. Which property determines the text that appears on a label control?

The Caption property of a label determines the text that is displayed on it.

Q.5. What is the purpose of Caption property of a command button and form?

Caption property of a command button indicates that text that appears on the command button. The Caption property of a form determines the text that appears in the form's title bar.

Q.6. What does cmdPush_Click mean?

cmdPush_Click is the name of an event procedure. It consists of object name cmdPush, an underscore and the name of event Click. It will be executed when a user clicks on the command button.

Q.7. Which property center-aligns the text in label? What should be the value of property?

Alignment property is used to center-align the text in a label. The value of the property should be "2-Center".

Q.8. When should you use a text box and when should you use a label control?

A text box should be used when input is required from the user or the user is allowed to change the text in textbox. A label should be used a message is displayed to the user. The user will not be able to change the text in a label.

Q.9. Check the following code. Correct the error if any.

```
Text1.Text = ITSeries
```

The correct syntax is:

```
Text1.Text = "ITSeries"
```

The text must be enclosed within the double quotation marks.

Q.10. How is it possible to enter data in multiple lines in a textbox?

A text box can accept and display multiple lines of text. It is done by setting the 'Multiline' property of the text box control to 'True'.

Q.11. How does an option button differ from check box?

In a group of check boxes, any number of boxes may be selected. Option buttons are used when only one button from a group may be selected.

Q.12. How can two groups of option buttons work independently?

Two groups of option buttons can work independently if each group is placed inside its own frame control. A group of option buttons inside a frame function together as a group.

Q.13. What is difference between 'Value' property of option button and Value property of checkbox?

The Value property of an option button is true if it is selected and false if it not selected. The Value property of checkbox is 0 if it not selected, 1 if it is selected and 2 if it is grayed.

Q.14. Explain the use of 'Default' and 'Cancel' properties of command buttons.

If the Default property of a command button is set to 'True', the user can press Enter instead of clicking the command button. Cancel button is the button that is selected when the user presses Esc key. If Cancel property of a button is set to true, the user can press Esp key instead of clicking that button.

Q.15. What is focus? How is possible to the focus of different objects?

In Windows programs, one control on form always has the 'focus'. If a button has the focus, a dotted line appears around the button. If a textbox has the focus, the cursor blinks in it.

Two properties determine whether the focus stops on a control and the order in which the focus moves. TabStop property can be set to 'True' or 'False'. If it is set to 'False', the focus will not stop on the control. TabIndex property determines the order of focus as the Tab key is pressed.

Q.16. What Basic statements will clear the current contents of a text box and a label?

The contents of a text box can be cleared by setting Text property to an empty string as follows:

```
Text1.Text = ""
```

The contents of a label can be cleared by setting Caption property to an empty string as follows:

```
Label1.Caption = ""
```

There should be no space between two quotation marks.

Q.17. Name and describe three styles of combo boxes.

The dropdown combo box (Style 0) has a dropdown list with text box. The list collapses when user makes selection from list. The simple combo box (Style 1) has a text box and the list displays below with scroll bars. The dropdown list (Style 2) is used to select from an existing list. It has no text box.

Q.18. How is possible to make scroll bars appear on a list box or combo box?

Both list boxes and combo boxes will add a scroll bar automatically if there are more items in the list than can be seen.

Q.19. When and how is information placed inside a list box or a combo box?

There are several ways to fill a list box or a combo box. The list items can be added at design time by using the List property. The list items can also be added at run time by using the AddItem method.

Q.20. Differentiate between linking and embedding an object in Visual Basic project.

Linking is used to access an object that is actually maintained by the application that creates it. A reference to the linked object is kept in code but the actual object is kept in the other application.

Embedding is used to place the object in code. Hence, the object is maintained from within the project and cannot be accessed by other projects.

Q.21. What is the purpose of the statement 'Unload Me'?

The keyword 'Me' refers to the form in which the code is written. Therefore, the statement 'Unload Me' unloads the current form from memory.

Q.22. What is the size of a twip?

The measurement system used in a project is known as scale. The default scale is twips. A twip is a twentieth of a point. A point is a measurement of 1/72nd of an inch and twip is 1/1440th of an inch.

Q.23. Which function loads a picture at run time?

The LoadPicture function is used in code to load a picture file at run time.

Q.24. How is a picture removed at run time?

A picture from the display can be removed at run time by setting the Visible property to False or by using the LoadPicture function with empty quotes.

Q.25. What is the purpose of the timer control?

Time control is used to execute a piece of code repeatedly after specified time interval. The Interval property is used to specify the interval. It can have a value of 0 to 65,535. It specifies the time interval in milliseconds.

Q.26. Explain how to make a graphic appear in an image control.

First place the image control on a form. Then select its Picture property in Properties window. You can set an image's Picture property to a file with an extension of .BMP, .ICO, .GIF, .JPG etc. The Stretch property can be set to True to make the picture enlarge according to the control size. The Visible property can be set to False to make the image disappear.

Q.27. What is the difference between List and ListIndex property?

ListIndex property is a number used to access individual elements in the list box. The list index starts with 0 for the first item in list box and takes on consecutive numbers for the remaining elements. The List property is used to return or set the items contained in the list box.

Multiple Choice

1. The main difference between a text box and a label is that:
 - a. User can modify text inside a text box but cannot modify the text in a label
 - b. User can modify text inside a label but cannot modify the text in a Textbox
 - c. No difference
 - d. Both a and b
2. Which one of the following is NOT a control?
 - a. Textbox
 - b. Close Button
 - c. Command Button
 - d. Option Button
3. Which of the following controls can be BEST used to display text as a heading?
 - a. Scroll bar
 - b. Optionbutton
 - c. Label
 - d. Shape

4. Which of the following is NOT a control?
a. PictureBox b. Command Button c. Menu d. ComboBox
5. The Interval property of Timer control is set in:
a. Minutes b. Hours c. Seconds d. Milliseconds
6. What is the value of 2880 Twips in inches?
a. 0.5 b. 1.0 c. 2.0 d. 4.0
7. Which of the following control is invisible at run time?
a. Label b. DriveListBox c. Timer d. Image
8. Which property is used to display multiple lines in a TextBox?
a. MaxLength b. Scrollbar c. Multiline d. None
9. Which event occurs when the user changes 'Text' property of a text box through a code?
a. Change b. ChangeText c. Both a, b d. None
10. Which code places the cursor in textbox txtName?
a. txtName.Focus b. GoTo txtName c. txtName.SetFocus d. txtName.Tab
11. To trigger the timer event once every second, set the interval property to:
a. 10 b. 100 c. 1000 d. 1
12. Which property returns the full location of a file stored on
a. AtWhere b. FileLoc c. Loc
13. In list box, which property is used to return the number of items
a. ListCount b. ListCounting c. ListOrder
14. Which property is used to access individual elements in the list box?
a. ListIndex b. ListCount c. ListOrder d. None
15. Which method is used to add the specified item to the list box?
a. Additem b. Sumitem c. Setfocus d. None
16. Which method is used to delete an item from the list box?
a. Additem b. Remove item c. Clearitem d. None
17. Which method is used to make the list box the current active element?
a. Setfocus b. LostFocus c. Active d. Unload
18. Which control combines the features of a text box and a list box?
a. Combo box b. Scrollbar c. Timer d. None
19. Which control is used to select one and only one of the multiple options:
a. Checkbox b. Radio option c. list d. Textbox
20. Which of the following is used in an application to give users true/false or yes/no options?
a. Checkbox b. Textbox c. List box d. None
21. Which property specifies the maximum number of characters that can be entered in textbox:
a. MaxLength b. MinLength c. Multiline d. None
22. A control that provides an identifiable grouping for controls is:
a. Frame Control b. Text Box c. Button d. None
23. A control that is used to display text or to accept user input is:
a. Label b. Text box c. Button d. None
24. 'Caption' property is NOT available with:
a. ListBox b. TextBox c. Both a and b d. None
25. When no item is currently selected in a ListBox, the 'ListIndex' property is:
a. 1 b. -1 c. 0 d. NULL
26. Which property prevents the control from responding to events when set to False?
a. Visible b. Caption c. Enabled d. Name

27. All items in a ListBox can be cleared at once using the method:
 - a. RemoveItem
 - b. Clear
 - c. Delete
 - d. None
28. To add a control at runtime, a control should be created at design time with 'Index' value of:
 - a. 0
 - b. 1
 - c. No value
 - d. 2
29. Which property is used to display the contents of combo box in alphabetical order?
 - a. Property
 - b. Alphabetical
 - c. Order
 - d. Sorted
30. Which object is used to identify drive and path that contain the current application?
 - a. App
 - b. Current
 - c. Form
 - d. Me
31. Which control is used to present the user with the possible options:
 - a. List box
 - b. Scrollbar
 - c. Textbox
 - d. None
32. Image controls hold a graphic that is assigned to:
 - a. Frame property
 - b. Text property
 - c. Caption property
 - d. Picture property
33. Which of the following is NOT a predefined object type in Visual Basic?
 - a. TextBox
 - b. CheckBox
 - c. Circle
 - d. ListBox
34. Setting the TabIndex of a control to 0 will :
 - a. display the control to the right side
 - b. Make the control Visible
 - c. Align the control to left side
 - d. Set the focus on it when program begins
35. Which of the following statements will clear the contents of a text box named txtMessage?
 - a. txtMessage.Text = ""
 - b. txtMessage.Caption = ""
 - c. lblMessage.Caption = ""
 - d. txtMessage.Text = "clear"
36. Which of the following instructions will add the word 'monkey' to list box lstAnimals?
 - a. lstAnimals.AddItem "monkey"
 - b. AddItem.lstAnimals monkey
 - c. lstAnimals.text = "monkey"
 - d. AddItem.lstAnimals "monkey"
37. Which of the following is NOT a style for combo boxes?
 - a. Simple Combobox
 - b. Dropdown combobox
 - c. Simple list
 - d. Drop downlist
38. How can you blank out an image?
 - a. Use LoadPicture function with a blank filename
 - b. Set Picture property of image to False
 - c. Set the Stretch property of the image to False
 - d. None
39. What property represents the current location of scroll box in scroll bar?
 - a. Min
 - b. Width
 - c. Index
 - d. Value
40. All of the following are examples of controls EXCEPT:
 - a. Label
 - b. Textbox
 - c. Property
 - d. Button
41. In the statement lblName.Text = "ImranTariq", lblName is a(n):
 - a. Method
 - b. Property
 - c. Object
 - d. Event
42. Entering text in a text box and then clicking a button can occur at:
 - a. Runtime
 - b. Debug time
 - c. Design time
 - d. Control time
43. Tabbing out of a text box is an example of a(n):
 - a. Control
 - b. Property
 - c. Event
 - d. Method
44. You can enter data in a control only when it has the:
 - a. Tab order
 - b. Mask
 - c. Focus
 - d. Control
45. The statement used to add "Accounting" to a list box named lstMajors at runtime is:
 - a. lstMajors.Items.Add("Accounting")
 - b. lstMajors.AddItem("Accounting")
 - c. lstMajors.Add("Accounting")
 - d. lstMajors.Add.Accounting
46. You have a list box named lstDepartments. The highest SelectedIndex value is:
 - a. 0
 - b. 1
 - c. lstDepartments.Items.Count
 - d. lstDepartments.Items.Count -1

47. To load a picture from computer's hard drive in a picture box at runtime:
 a. Use Image. Load method b. Use Image.FromFile method
 c. Set Image property in the Properties window d. Set Image property to Nothing
48. Given a label with the text EmployeeID, when the user presses Alt-I:
 a. Focus moves to the label b. Focus moves to a control after label in tab order
 c. Focus moves to a control next to label d. Nothing will happen
49. Which method is used to locate an item in a list box?
 a. SetSelected b. Selected c. Index d. SetIndex
50. What is the major difference between a ListBox and ComboBox?
 a. ComboBox displays a list of items
 b. ListBox does not include TextBox for data entry
 c. A ListBox includes a TextBox for data entry
 d. None
51. The _____ method causes the control to be redrawn:
 a. SetFocus b. Refresh c. Visible d. Move
52. How many options can be selected if a frame contains four option buttons?
 a. 1 b. 2 c. 3 d. 0

Answers

1. a	2. b	3. c	4. c
5. d	6. c	7. c	8. c
9. a	10. c	11. c	12. d
13. a	14. a	15. a	16. b
17. a	18. a	19. b	20. a
21. a	22. a	23. b	24. c
25. b	26. c	27. b	28. a
29. d	30. a	31. a	32. d
33. c	34. d	35. a	36. a
37. c	38. a	39. c	40. c
41. c	42. a	43. c	44. c
45. a	46. d	47. c	48. b
49. a	50. b	51. b	52. a

Fill in the Blanks

- Visual Basic provides two types of buttons: command buttons and _____ buttons.
- A _____ is used to identify controls that do not have a caption of their own.
- _____ provide easy navigation through a long list of items or a large amount of information.
- A _____ control provides an identifiable grouping for other controls.
- _____ boxes display long lists of options from which users can choose.
- The _____ property of a control determines the order in which it receives the focus.
- A _____ represents a data value stored inside or associated with an object or control.
- For objects that may receive user input, _____ property determines whether or not the objects can receive the focus when the user presses the Tab key.

9. _____ property of a label allows the caption to be displayed as left or right justified.
10. To delete an item from a list box, use the _____ method.
11. To display the contents of a combo box in alphabetical order, set _____ property to true.
12. Checkbox is an example of a _____ object type.
13. To reference a control in code, use the control's _____ property.
14. A group of controls having the same name is called a _____.
15. Radio button controls are commonly called _____ buttons.
16. A control that is used to display a simple message is a(n) _____.
17. To make a control inaccessible to a user, set its _____ property to _____.
18. The _____ event is the only event of the Timer control.
19. To clear the contents of a text box, set its Text property to a(n) _____ string.
20. A control with its Enabled property set to _____ cannot receive focus.
21. There are three radio buttons in a group box. The Checked property can be True for a maximum of _____ of the buttons.
22. The first item in a list box has a _____ value of 0.
23. JPG stands for _____.
24. BMP stands for _____.
25. The _____ property indicates the current directory path.
26. The _____ property returns the name of the file selected from the file list control.
27. The _____ control shows a list of filenames of the selected folder.
28. The _____ event is triggered whenever the drive selection is changed.
29. OLE stands for _____.

Answers

1. Image	2. Label	3. Scrollbar
4. Frame	5. List	6. TabIndex
7. Property	8. Tabstop	9. Alignment
10. Removeitem	11. Sorted	12. Predefined
13. New	14. Control array	15. Option
16. Label	17. Enabled, False	18. Tick
19. Zero-length	20. False	21. One
22. Selected index	23. Joint photograph Exports Graph	24. Bitmap
25. Drive	26. Filename	27. Pattern
28. Change	29. Object linking & embedding	

True / False

1. Controls are reusable, predefined components used for visual programming.
2. Controls serve as building blocks that can be quickly combined to create applications.
3. Visual Basic controls exist in two varieties intrinsic controls and ActiveX controls.
4. Intrinsic controls are the default controls provided in Visual Basic toolbox.
5. Text boxes can display text but cannot input text.
6. The Text property of a control can be left blank.

7. Changing the Name property of a form will have a visible effect when the application is run.
8. The value of TabStop property determines the order in which a control receives the focus.
9. Option buttons are usually grouped or set apart by using the Line control.
10. A single name can be assigned to a group of controls of the same type.
11. Copies of a control are assigned the properties of the first instance of the control.
12. Compare Picture properties of two image controls to determine if image controls are identical.
13. Additional controls may be purchased from third-party vendors for use with Visual Basic programs.
14. The TabStop property indicates whether a control can be tabbed to by the user.
15. Label control is used to have a user enter text onto a form.
16. A listbox control has a Clear method.
17. A picture box can display both text and graphics.
18. Both text boxes and command buttons have a click event.
19. All controls have a TabStop.
20. Once a PictureBox is defined, its size cannot be modified.
21. The Name property of a control cannot be left blank.

Answers

1. T	2. T	3. T	4. T
5. F	6. F	7. F	8. F
9. F	10. T	11. T	12. F
13. T	14. T	15. F	16. T
17. T	18. T	19. F	20. F
21. T			

Project 4

Students Information

Project Specification

The project **Students Information** displays a form that uses textboxes in a frame to input Name, Phone, Address and Email.

It uses another frame to display two radio buttons to select gender. It displays another frame to display two radio buttons to select education level. It displays one combo box to display the list of undergraduate courses and second combo box to display the list of postgraduate categories. When the user selects a category from combo box, the detail of courses of that category is displayed in a list box. The user can select multiple courses from the list. A button calculates total fee for the selected courses.

Procedure

1. Start a new Standard EXE project.
2. Place the following controls on the form with the given properties:

Control	Control Name	Property	Value
Frame	frmInfo	Caption	Student Information
Label	lblName	Caption	Name
Label	lblPhone	Caption	Phone
Label	lblAddress	Caption	Address
Label	lblEmail	Caption	Email
Textbox	txtName	Text	Blank
Textbox	txtPhone	Text	Blank
Textbox	txtAddress	Text	Blank
Textbox	txtEmail	Text	Blank
Frame	frmGender	Caption	Gender
OptionButton	optMale	Caption	Male
OptionButton	optFemale	Caption	Female
Frame	frmEducation	Caption	Education Level
OptionButton	optUnder	Caption	Undergraduate
OptionButton	optPost	Caption	Postgraduate
ComboBox	cmbUnder	Style	2 - Dropdown List
		List	Bcom BBA BCS
ComboBox	cmbPost	Style	2 - Dropdown List
		List	MA MSc
Listbox	lstSub	Style	1 - Checkbox
CommandButton	cmdCalculate	Caption	Calculate Fee

3. Design the form as follows:

4. Add the following code in Click event of optUnder radio button:

```

Project1 - Form1 (Code)
optUnder Click
Private Sub optUnder_Click()
    cmbUnder.Enabled = True
    cmbPost.Enabled = False
End Sub

```

5. Add the following code in Click event of optPost radio button:

```

Project1 - Form1 (Code)
optPost Click
Private Sub optPost_Click()
    cmbPost.Enabled = True
    cmbUnder.Enabled = False
End Sub

```

6. Add the following code in Click event of cmbUnder combo box:

```

Project1 - Form1 (Code)
cmbUnder Click
Private Sub cmbUnder_Click()
    Dim u As String
    Dim BCom(4) As String
    Dim BBA(4) As String
    Dim BS(4) As String
    BCom(0) = "English"
    BCom(1) = "Accounting"
    BCom(2) = "Economics"
    BCom(3) = "Business Math"
    BCom(4) = "Business Communication"
    BBA(0) = "Introduction to Business"
    BBA(1) = "Accounting"
    BBA(2) = "Financial Management"
    BBA(3) = "Business Communication"
    BBA(4) = "Introduction to IT"
    BS(0) = "Introduction to Programming"
    BS(1) = "Introduction to IT"
    BS(2) = "Digital Electronics"
    BS(3) = "Mathematics"
    BS(4) = "Introduction to Databases"
    u = cmbUnder.Text
    If u = "BCom" Then
        1stSub.Clear
    End If
End Sub

```



```

For i = 0 To 4
    lstSub.AddItem BCom(i)
Next
ElseIf u = "BBA" Then
    lstSub.Clear
    For i = 0 To 4
        lstSub.AddItem BBA(i)
    Next
Else
    lstSub.Clear
    For i = 0 To 4
        lstSub.AddItem BS(i)
    Next
End If
End Sub

```

7. Add the following code in Click event of cmbPost combo box:

Project1 - Form1 (Code)

cmbPost Click

```

Private Sub cmbPost_Click()
    Dim u As String
    Dim MA(2) As String
    Dim MSc(2) As String
    MA(0) = "English"
    MA(1) = "Urdu"
    MA(2) = "Economics"
    MSc(0) = "Physics"
    MSc(1) = "Chemistry"
    MSc(2) = "Biology"
    u = cmbPost.Text
    If u = "MA" Then
        lstSub.Clear
        For i = 0 To 2
            lstSub.AddItem MA(i)
        Next
    Else
        lstSub.Clear
        For i = 0 To 2
            lstSub.AddItem MSc(i)
        Next
    End If
End Sub

```

8. Add the following code in Click event of cmdCalculate button:

Project1 - Form1 (Code)

cmdCalculate Click

```

Private Sub cmdCalculate_Click()
    Dim n As Integer, fee As Integer
    If cmbUnder.Enabled Then
        n = 0
        For i = 0 To lstSub.ListCount - 1
            If lstSub.Selected(i) Then n = n + 1
        Next
        fee = n * 2000
        MsgBox ("Your total fee is " & fee)
    ElseIf cmbPost.Enabled Then
        MsgBox ("Contact office for postgraduate fee.")
    End If
End Sub

```


9. Run the project and enter values as follows:

Student Information

Student Information

Name Phone

Address

Email

Gender
☒ Male ☐ Female

Education Level
☒ Undergraduate ☐ Postgraduate

BCS

☒ Introduction to Programming
☒ Introduction to IT
☐ Digital Electronics
☐ Mathematics
☒ Introduction to Databases

Calculate Fee

10. Click **Calculate Fee** button. The result will appear as follows:

